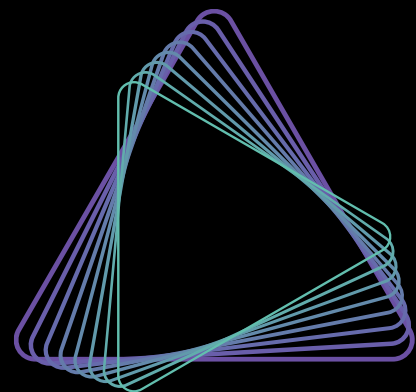




**AZENTA 安升达**  
LIFE SCIENCES

# 单细胞基因编辑报告解读及 数据分析

2024年08月

# 目录

|                |    |
|----------------|----|
| 分析所需软件         | 01 |
| 什么是scCRISPR    | 02 |
| 怎么分析scCRISPR数据 | 03 |

# 分析所需软件

01



# 1 分析所需软件

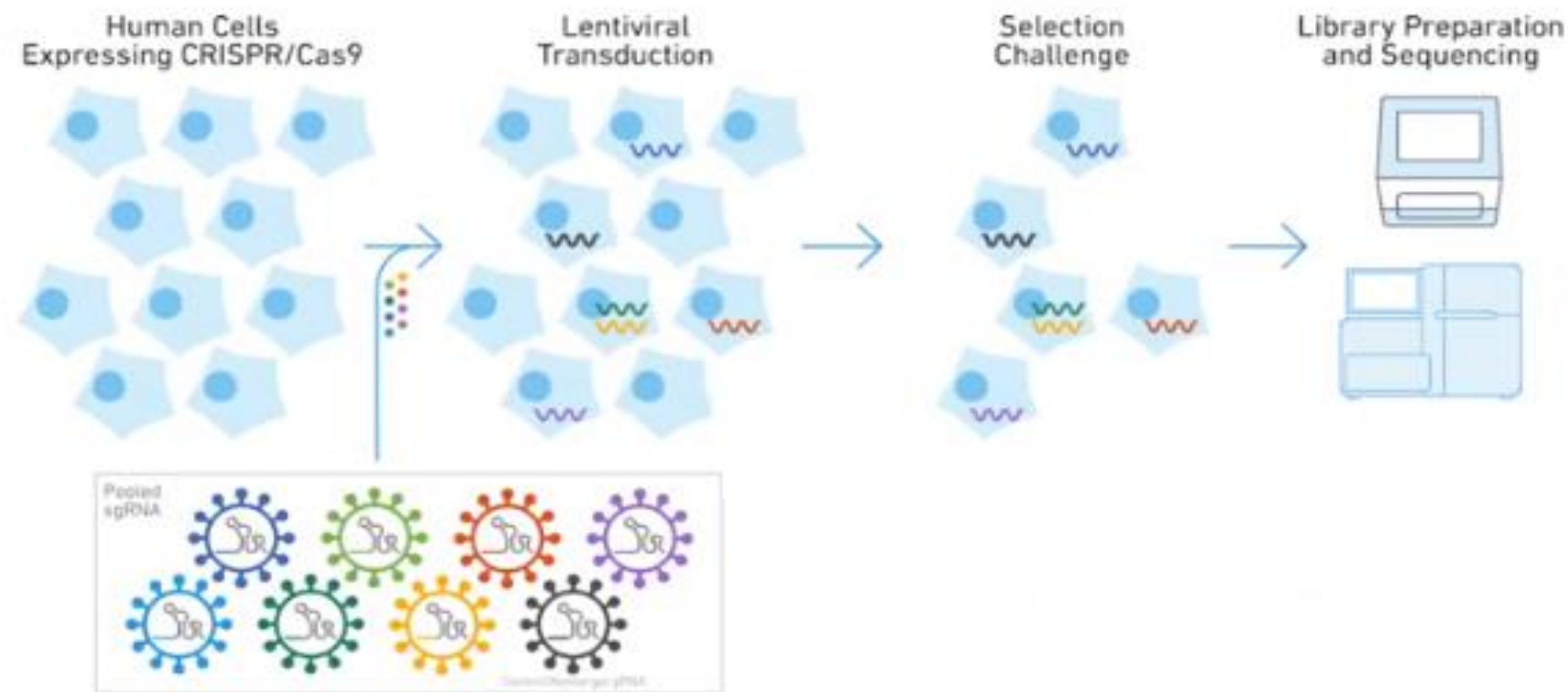
- cellranger
- R (包括Seurat、ggplot2等包)
- python (包括sys、os等包)
- gzip

# 什么是scCRISPR

02



## 2 什么是scCRISPR



单细胞CRISPR是一种基因编辑和分析方法，旨在通过在单个细胞中引入CRISPR系统（如CRISPR-Cas9），并利用单细胞测序技术（如单细胞RNA测序、单细胞DNA测序等）来研究和评估基因编辑的效果。

这种方法允许研究人员在单细胞水平上观察基因编辑的精确结果、细胞间的异质性以及编辑后的表型变化。

# 怎么分析scCRISPR数据

# 03

# 3.1 分析前准备 --- cellranger



下载地址：<https://www.10xgenomics.com/support/software/cell-ranger/downloads>

### End User Software License for 10x Genomics software products.

PLEASE CAREFULLY REVIEW ALL OF THE TERMS SET FORTH IN THIS LICENSE AGREEMENT BEFORE CLICKING "I AGREE", SIGNING OR OTHERWISE INSTALLING, USING, OR ACCESSING THE 10x GENOMICS SOFTWARE PRODUCT, AS BY DOING SO, YOU ("LICENSEE") ARE ACKNOWLEDGING THAT YOU HAVE READ AND UNDERSTAND THIS LICENSE AGREEMENT AND ARE AGREEING TO BE BOUND BY ALL OF ITS TERMS.

#### 1.0 General

1.1 This agreement ("Agreement") relates to the license, installation and/or use by you (the "Licensee") of the software product(s) of 10x Genomics, Inc. ("10x"), identified herein and/or that accompanies these license terms, including any and all associated modules, documentation, and updates or revisions thereto.

1.2 By agreeing and/or otherwise proceeding with the installation of, use of or other access to the 10x software products, you acknowledge that you have read and understood all of the terms of this Agreement, and that you are agreeing to be bound by all of the terms set forth herein. 10x does not accept, and expressly rejects hereby any conflicting or additional terms presented by Licensee, and such terms shall have no effect, whether or not 10x communicates the rejection again, specifically after receipt of licensee's terms. The Agreement consists only of these terms and any other terms that 10x issues or expressly accepts in a duly signed writing. Any additional software provided by 10x may be subject to additional and/or different terms and conditions.

#### 2.0 Definitions

2.1 "Product" means each item provided by 10x to Licensee, including any 10x Instrument, Licensed Software and/or 10x Consumable, listed in the applicable written sales quotation issued by 10x ("Quotation") or, in the event a Quotation is not issued, listed in 10x's then-current price list applicable to Buyer's jurisdiction, and identified in Buyer's purchase order.

2.2 "10x Instrument" means a 10x-branded instrument sold by 10x or an authorized distributor and/or reseller.

2.3 "10x Consumables" means 10x microfluidic cartridges and/or associated 10x-branded reagent kits, and labware, or any individual components of the foregoing, intended by 10x for use with 10x Instruments.

### Download Cell Ranger

Before you download 10x software, please agree to the End User License Agreement.

First Name

Last Name

Email

Institution

Country/Region

Industry

Area of Research

How can we help you?

☐ I would like to receive communications from 10x Genomics. *Note: you may change your email preferences at a later date.*



# 3.2 参考基因组下载



下载地址：  
<https://www.10xgenomics.com/support/software/cell-ranger/downloads#reference-downloads>

目前10X只提供人与小鼠的参考基因组

自定义基因组需使用 cellranger mkref

[Download 2020-A references >](#)

## Human reference (GRCh38) - 2024-A

- Human reference (GRCh38) dataset required for Cell Ranger.

[Download Human Reference](#)

File size: 11 GB

md5sum: a7b5b7ceefe10e435719edc1a8b8b2fa

[curl](#) [wget](#)

[Copy](#)

```
curl -O "https://cf.10xgenomics.com/supp/cell-exp/refdata-gex-GRCh38-2024-A.tar.gz"
```

## Mouse reference (GRCm39) - 2024-A

- Mouse reference (GRCm39) dataset required for Cell Ranger.

[Download Mouse Reference](#)

File size: 9.7 GB

md5sum: 37c51137ccaeabd4d151f80dc86ce0b3

## 3.3.1 cellranger count 命令

```
cellranger count \  
--id=scCRISPR \  
--fastqs=mRNA \  
--sample=scCRISPR \  
--transcriptome=refdata-gex-mm10-2020-A
```

普通单细胞转录组count命令

```
cellranger count \  
--id=scCRISPR \  
--libraries=lib.csv \  
--feature-ref=feature.csv \  
--transcriptome=refdata-gex-mm10-2020-A
```

CRISPR单细胞转录组count命令

# 3.3.2 scCRISPR 所需配置文件之lib.csv

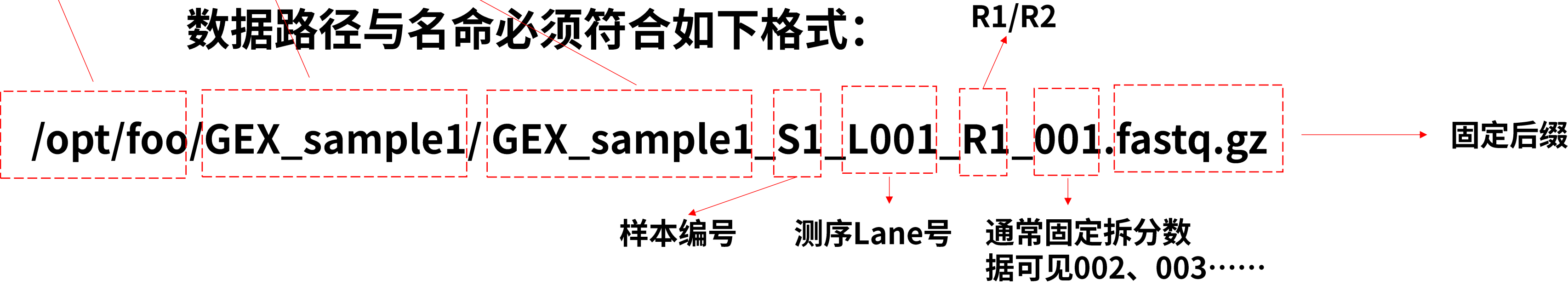


**Gene expression + CRISPR Guide Capture libraries.** In this example, we have demultiplexed sequencing data from two libraries named GEX\_sample1 and CRISPR\_sample1 on the `bc12fastq / mkfastq` sample sheet. This generated two FASTQ files named `GEX_sample1_S0_L001_R1_001.fastq.gz` and `CRISPR_sample1_S0_L001_R1_001.fastq.gz` in the path `/opt/foo` (be sure to use the correct full path to your FASTQ files). We pass the FASTQ sample names and paths to Cell Ranger with the appropriate library types:

```
fastqs,sample,library_type,  
/opt/foo/,GEX_sample1,Gene Expression,  
/opt/foo/,CRISPR_sample1,CRISPR Guide Capture,
```

数据存放路径      样本名称      样本类型（根据实验设计，选填“Gene Expression”或者“CRISPR Guide Capture”）

数据路径与名命必须符合如下格式：





# 3.3.3 scCRISPR 所需配置文件之feature.csv



## CRISPR Guide Capture

In CRISPR Guide Capture assays, the Feature Barcode sequence is the [CRISPR protospacer](#) sequence. The protospacer is followed by a downstream constant sequence in the guide RNA which is used as an anchor to identify the location of the protospacer. We recommend using a 12bp-20bp constant sequence that can be uniquely identified but is short enough that it is unlikely to be disrupted by a sequencing error.

The example Feature Reference CSV files list six guide RNA features, each with six distinct barcode/protospacer sequences (`sequence` column). The `pattern` column has the same pattern for all six features. Please note that the `pattern` field is specific to your experiment and must be customized. Remember to replace the example `pattern` sequence provided here when running your own analysis.

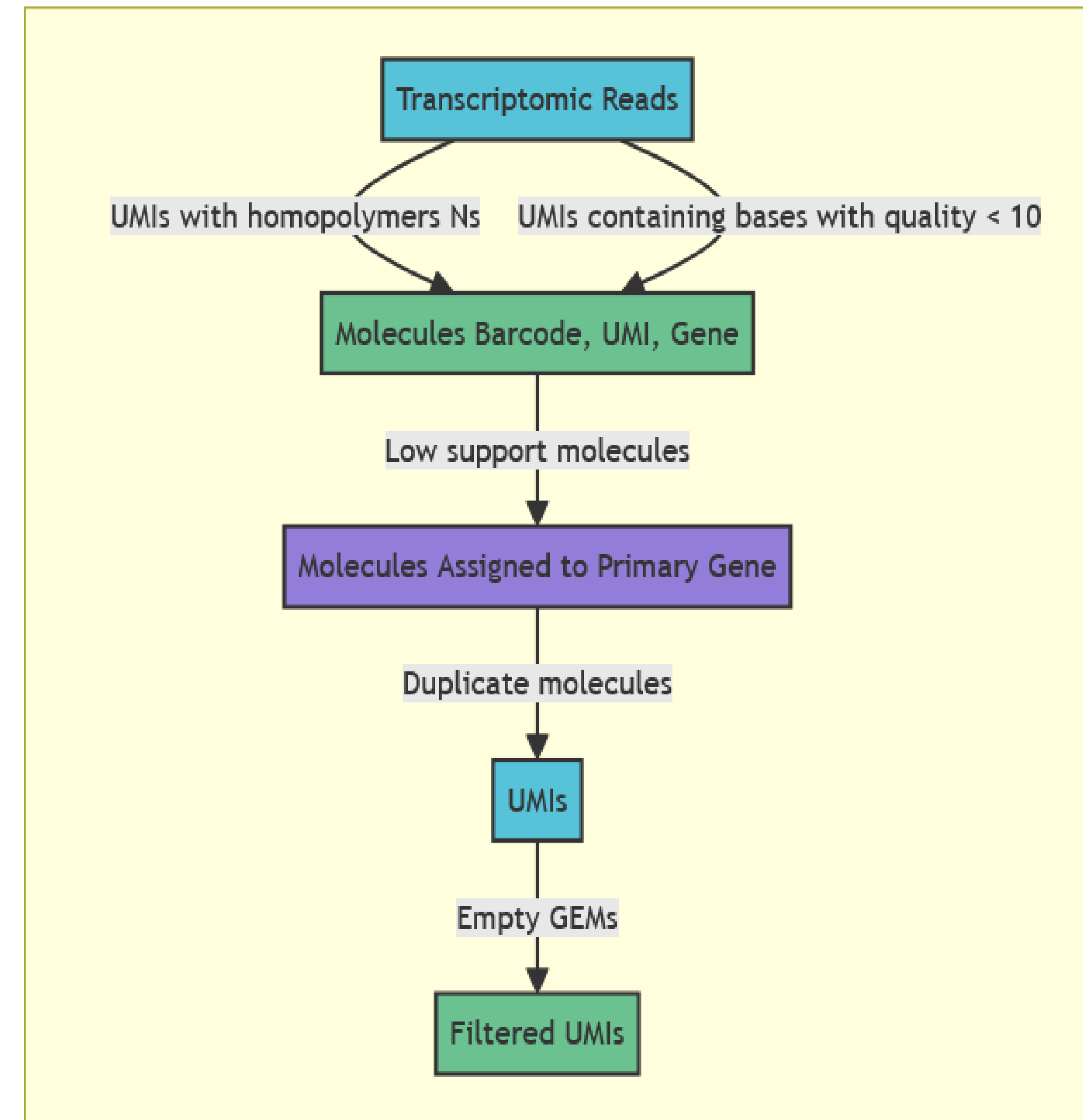
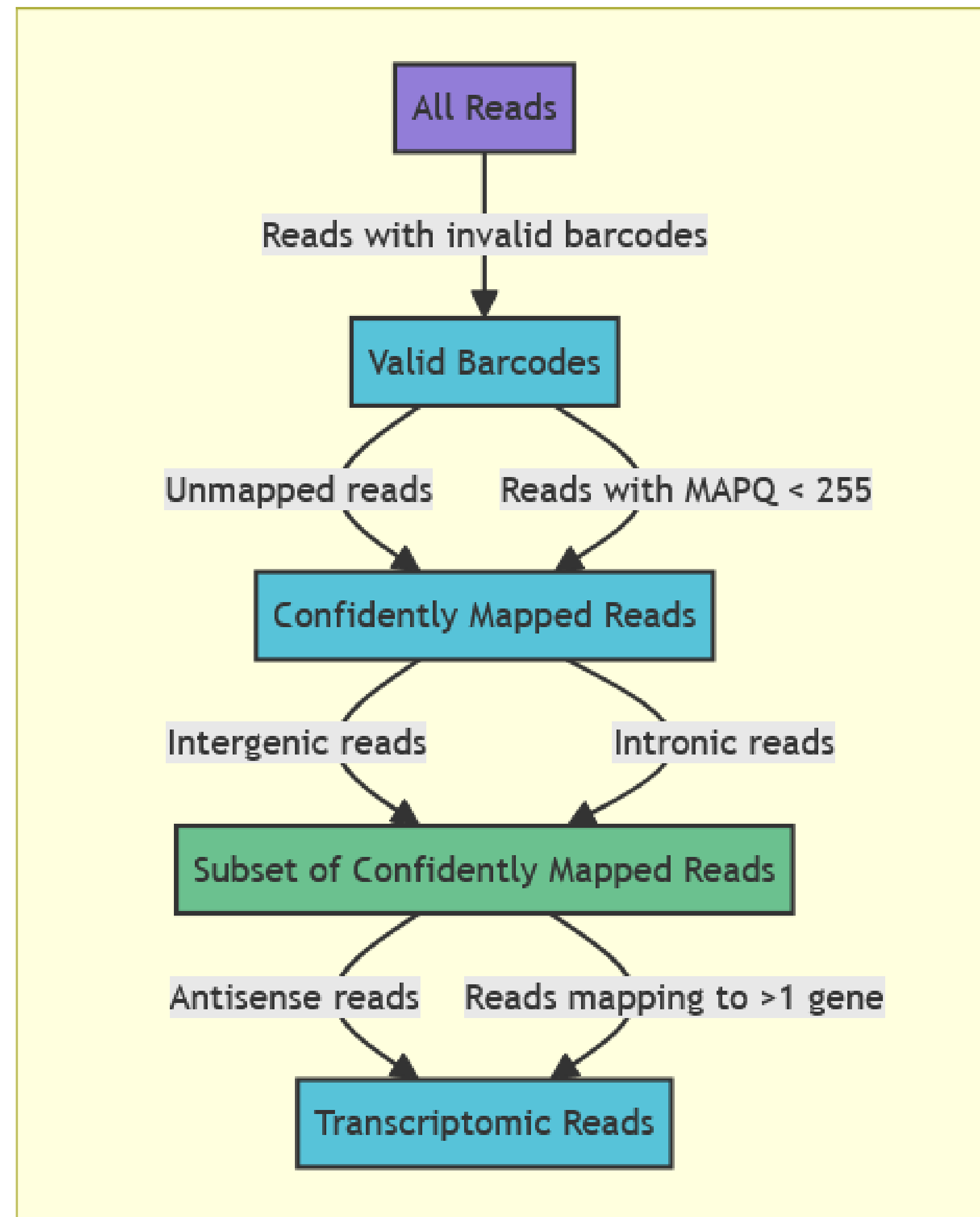
We use the `target_gene_id` and `target_gene_name` columns to declare the target gene of each guide RNA, for use in downstream CRISPR perturbation analysis. Two guides are declared with `target_gene_id` as `Non-Targeting`. Cells containing `Non-Targeting` guides will be used as controls for CRISPR perturbation analysis. The four remaining guides target two genes.

| Read | Pattern                  | Assay                                   | Example                         |
|------|--------------------------|-----------------------------------------|---------------------------------|
| R2   | (BC)GTTTAAGAGCTAAGCTGGAA | 3' Gene Expression with Feature Barcode | <a href="#">Download 3' CSV</a> |
| R2   | TTCCAGCATAGCTCTTAAAC(BC) | 5' Gene Expression with Feature Barcode | <a href="#">Download 5' CSV</a> |

gRNA 唯一ID      根据sgRNA所在位置选择R1/R2      gRNA 序列（不可重复）  
根据捕获方式选则 “(BC)固定序列” 或者 “固定序列(BC)”

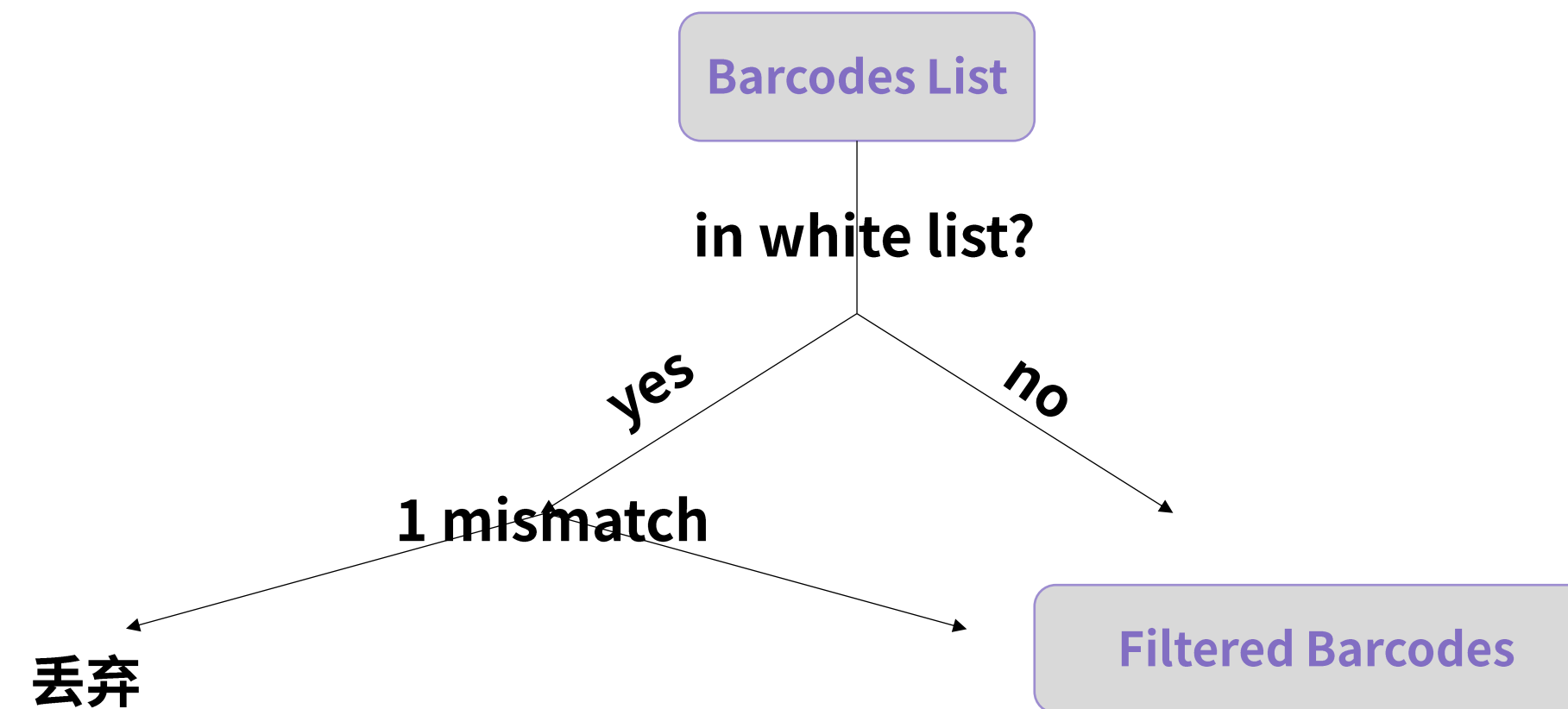
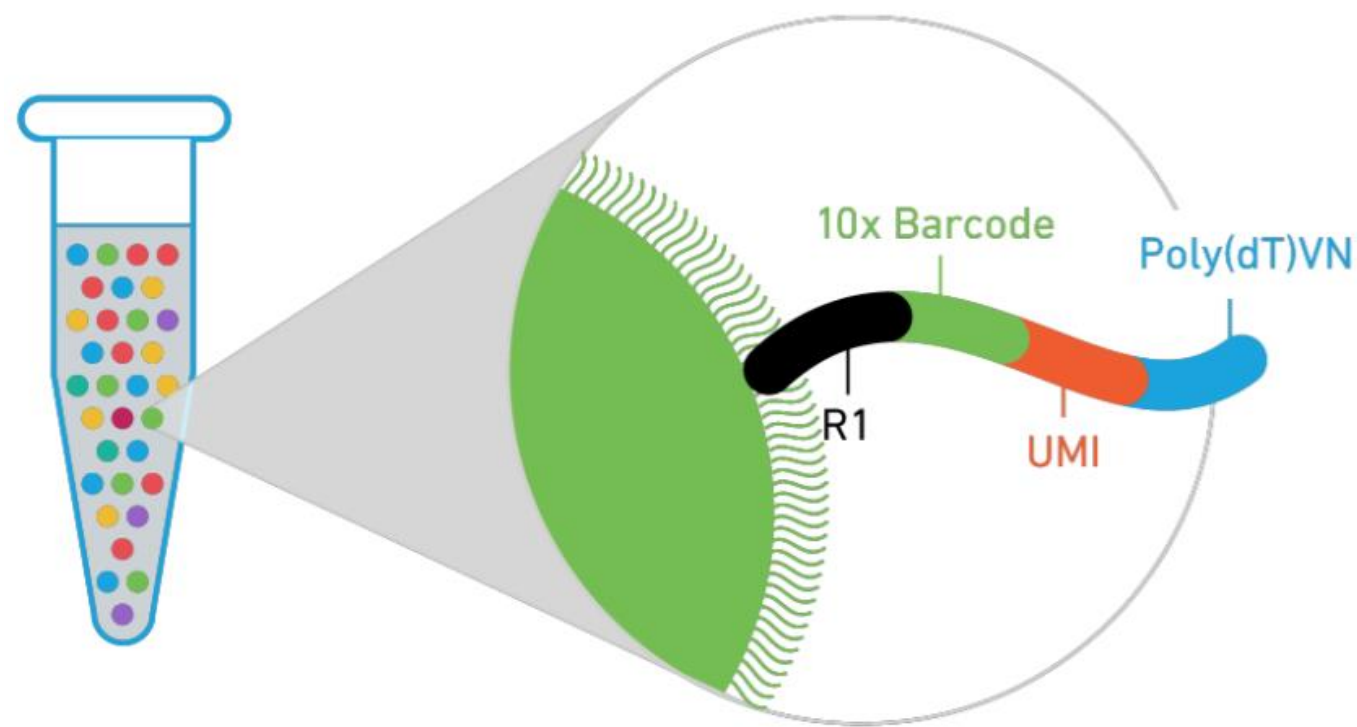
```
id,name,read,pattern,sequence,feature_type
gRNA1,gRNA1,R2,GCTATTTCTAGCTCTAAAAC(BC),ATGGCGAGTGATCCCTCACG,CRISPR Guide Capture
gRNA2,gRNA2,R2,GCTATTTCTAGCTCTAAAAC(BC),CAGTATCGGGGGTTCAGAGC,CRISPR Guide Capture
gRNA3,gRNA3,R2,GCTATTTCTAGCTCTAAAAC(BC),ATACTGAGGGGTGGCAAACA,CRISPR Guide Capture
gRNA4,gRNA4,R2,GCTATTTCTAGCTCTAAAAC(BC),CTGGCAGGCGCGACTCCCTG,CRISPR Guide Capture
gRNA5,gRNA5,R2,GCTATTTCTAGCTCTAAAAC(BC),TAATGGGCAGCTTGCGGCTG,CRISPR Guide Capture
gRNA6,gRNA6,R2,GCTATTTCTAGCTCTAAAAC(BC),TTAAGGGAGCGGAACACATG,CRISPR Guide Capture
gRNA7,gRNA7,R2,GCTATTTCTAGCTCTAAAAC(BC),GTCGTAGTCTCGATATTCAC,CRISPR Guide Capture
gRNA8,gRNA8,R2,GCTATTTCTAGCTCTAAAAC(BC),CGACACGAACTAAGCCCCC,CRISPR Guide Capture
gRNA9,gRNA9,R2,GCTATTTCTAGCTCTAAAAC(BC),GATGAGGTGGGAAAGCGTCG,CRISPR Guide Capture
gRNA10,gRNA10,R2,GCTATTTCTAGCTCTAAAAC(BC),TGCGCGCTAAACAGCTGGAC,CRISPR Guide Capture
gRNA11,gRNA11,R2,GCTATTTCTAGCTCTAAAAC(BC),GCTGTTTAGCGCGCAAGGCG,CRISPR Guide Capture
gRNA12,gRNA12,R2,GCTATTTCTAGCTCTAAAAC(BC),CATGGACTCTAACTACTTGG,CRISPR Guide Capture
gRNA13,gRNA13,R2,GCTATTTCTAGCTCTAAAAC(BC),AGGTTATATTCTCGTAGTCA,CRISPR Guide Capture
gRNA14,gRNA14,R2,GCTATTTCTAGCTCTAAAAC(BC),GACTACGAGAATATAACCTT,CRISPR Guide Capture
gRNA15,gRNA15,R2,GCTATTTCTAGCTCTAAAAC(BC),GCACAGTCTTCAGCCGGTCG,CRISPR Guide Capture
gRNA16,gRNA16,R2,GCTATTTCTAGCTCTAAAAC(BC),AAAGATCTGGCGGTGCCACC,CRISPR Guide Capture
gRNA17,gRNA17,R2,GCTATTTCTAGCTCTAAAAC(BC),TTGTTCGCCTTCAAGGTAGG,CRISPR Guide Capture
gRNA18,gRNA18,R2,GCTATTTCTAGCTCTAAAAC(BC),TGAAGGCGAACAATGGCTGG,CRISPR Guide Capture
gRNA19,gRNA19,R2,GCTATTTCTAGCTCTAAAAC(BC),GGTGTAGTACTCGCTTCGAT,CRISPR Guide Capture
gRNA20,gRNA20,R2,GCTATTTCTAGCTCTAAAAC(BC),AGTATCGAAGCGAGTACTAC,CRISPR Guide Capture
```

### 3.3.4 cellranger count 数据处理流程





### 3.3.5 cellranger count 之 barcode 矫正



- Gel beads在设计时一个磁珠上的barcode是完全相同的
- 一管beads所包含的所有barcode种类 (white list) 是已知的

观察barcode是否在白名单中，如果不在，最多允许一个错配去匹配，若能匹配到则回收，反之抛弃。

# 3.3.6 cellranger count 之比对

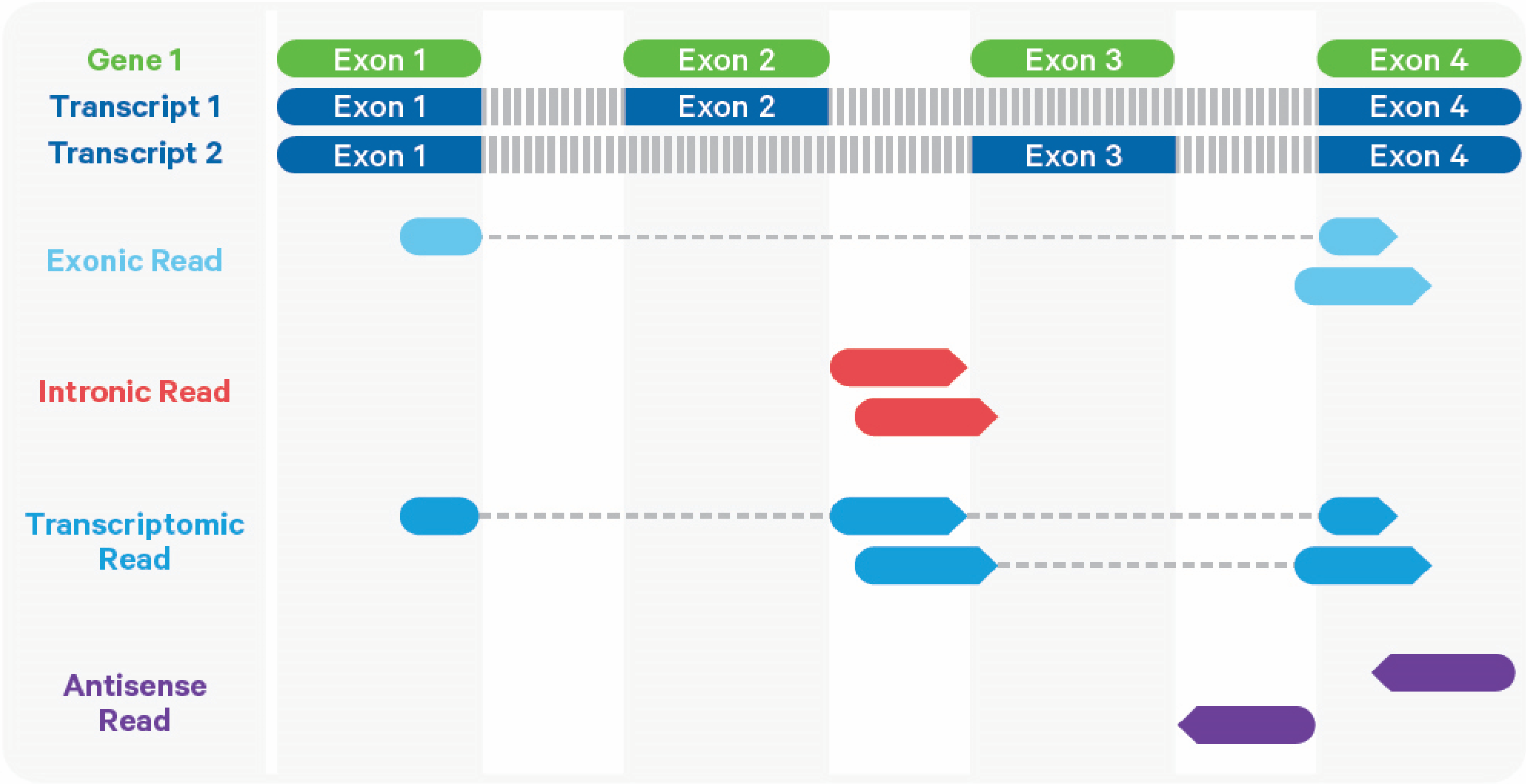
使用STAR进行比对

根据注释文件将参考基因组分为三类-外显子、内含子和基因间区。

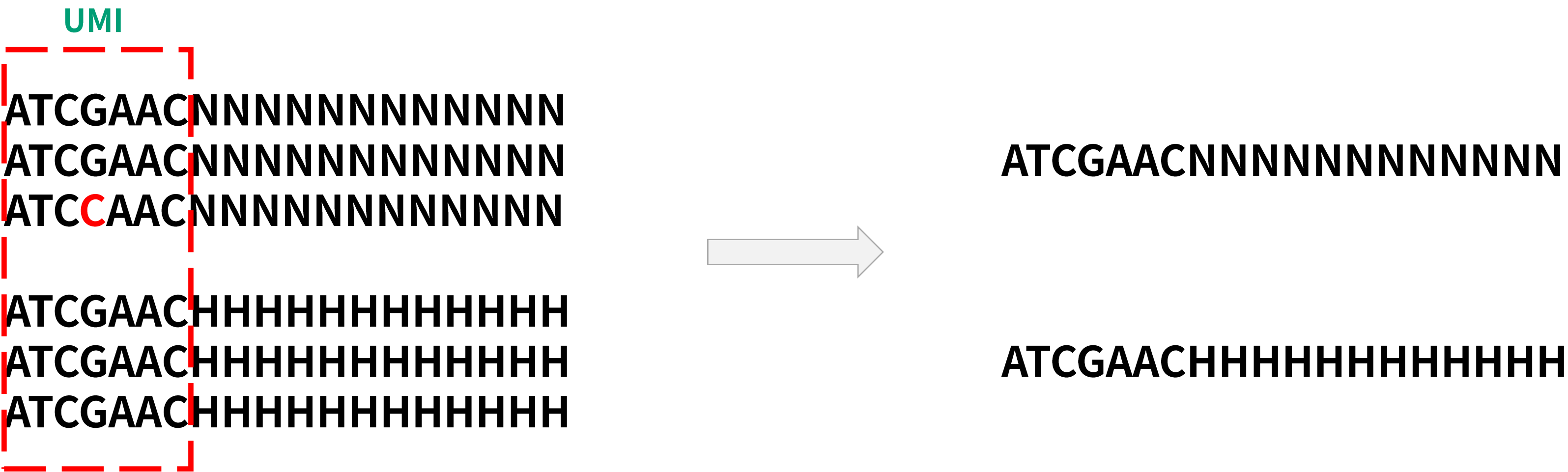
reads与外显子overlap≥50%则为外显子；

若非外显子reads与内含子存在overlap则为内含子；

其余为基因间区reads；



# 3.3.7 cellranger count 之 UMI 矫正

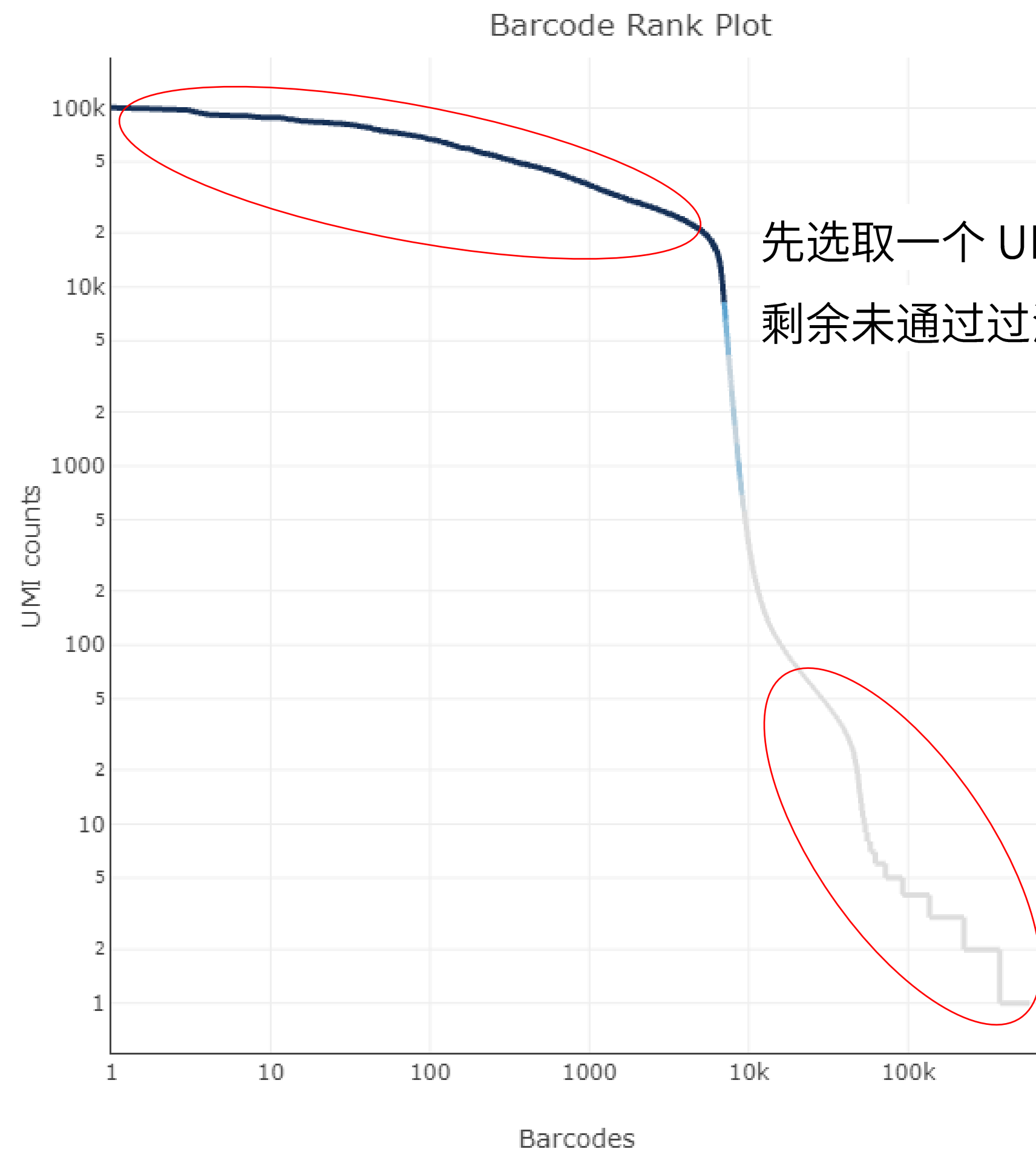


比对位置相同、UMI相同的情况下，这组序列会被认为是来源于同一个分子，计数为1

两种UMI只相差一个碱基不一致的情况下，少数服从多数，依然认为是来源于同一个分子



# 3.3.8 cellranger count 之细胞鉴定



- UMI 去重校正后，每一个cell barcode对应一个候选细胞
- 每个候选细胞中包含若干UMI标记的reads

先选取一个 UMI 总数的阈值，UMI总数大于这个阈值的候选细胞被识别为细胞。  
剩余未通过过滤的候选细胞将与背景表达谱进行比较。

选则UMI数极低，但包含较多候选细胞的阈值，UMI数低于该阈值的候选细胞标记为背景，  
根据其表达特征建模。

上一步未被标记为真细胞的候选细胞将与该背景模型比较，差异显著的候选细胞标记为真细胞。

# 3.3.9 cellranger count 结果报告



Summary

Analysis

20,000

Estimated Number of Cells

36,256

Mean Reads per Cell

4,000

Median Genes per Cell

Sequencing

|                               |             |
|-------------------------------|-------------|
| Number of Reads               | 779,023,469 |
| Number of Short Reads Skipped | 0           |
| Valid Barcodes                | 91.0%       |
| Valid UMIs                    | 99.8%       |
| Sequencing Saturation         | 28.7%       |
| Q30 Bases in Barcode          | 96.7%       |
| Q30 Bases in RNA Read         | 90.1%       |
| Q30 Bases in RNA Read 2       | 90.7%       |
| Q30 Bases in UMI              | 96.7%       |

Cells

Barcode Rank Plot

|                            |        |
|----------------------------|--------|
| Estimated Number of Cells  | 20,000 |
| Fraction Reads in Cells    | 88.3%  |
| Mean Reads per Cell        | 36,256 |
| Median Genes per Cell      | 4,000  |
| Total Genes Detected       | 18,932 |
| Median UMI Counts per Cell | 14,230 |

Mapping

|                                                |       |
|------------------------------------------------|-------|
| Reads Mapped to Genome                         | 89.6% |
| Reads Mapped Confidently to Genome             | 86.7% |
| Reads Mapped Confidently to Intergenic Regions | 5.5%  |
| Reads Mapped Confidently to Intronic Regions   | 5.8%  |
| Reads Mapped Confidently to Exonic Regions     | 75.5% |
| Reads Mapped Confidently to Transcriptome      | 68.1% |
| Reads Mapped Antisense to Gene                 | 5.4%  |

CRISPR Sequencing

|                               |            |
|-------------------------------|------------|
| Number of Reads               | 69,328,236 |
| Number of Short Reads Skipped | 0          |
| Valid Barcodes                | 97.5%      |
| Valid UMIs                    | 100.0%     |
| Sequencing Saturation         | 64.0%      |
| Q30 Bases in Barcode          | 96.9%      |
| Q30 Bases in CRISPR Read      | 90.3%      |
| Q30 Bases in CRISPR Read 2    | 94.6%      |
| Q30 Bases in UMI              | 96.3%      |

Sample

|                    |                                         |
|--------------------|-----------------------------------------|
| Sample ID          | scCRISPR                                |
| Sample Description |                                         |
| Chemistry          | Single Cell 5' PE                       |
| Include introns    | False                                   |
| Reference Path     | ...nce/02.Mouse/refdata-gex-mm10-2020-A |
| Transcriptome      | mm10-2020-A                             |
| Pipeline Version   | cellranger-6.0.1                        |

CRISPR Application

|                                                   |       |
|---------------------------------------------------|-------|
| Fraction Reads with Putative Protospacer Sequence | 90.6% |
| Fraction Guide Reads                              | 87.0% |
| Fraction Guide Reads Usable                       | 67.5% |
| Guide Reads Usable per Cell                       | 2,177 |
| Fraction Protospacer Not Recognized               | 3.9%  |
| Guide Reads in Cells                              | 79.3% |
| Cells with 1 or more protospacers detected        | 84.8% |
| Cells with 2 or more protospacers detected        | 22.5% |
| Median UMIs per Cell                              | 320   |



# 3.3.10 cellranger count 结果文件



outs 文件夹

```
2024 analysis
2024 cloupe.cloupe
2024 crispr_analysis
2024 feature_reference.csv
2024 filtered_feature_bc_matrix
2024 filtered_feature_bc_matrix.h5
2024 metrics_summary.csv
2024 molecule_info.h5
2024 possorted_genome_bam.bam
2024 possorted_genome_bam.bam.bai
2024 raw_feature_bc_matrix
2024 raw_feature_bc_matrix.h5
2024 web_summary.html
```

barcodes.tsv.gz

```
AAACCTGAGGAATTAC-1
AAACCTGAGGAGCGTT-1
AAACCTGAGGCTAGGT-1
AAACCTGAGGGATCTG-1
AAACCTGAGTACGTTC-1
AAACCTGCAAACGCGA-1
AAACCTGGTAAACCTC-1
AAACCTGGTACCGAGA-1
AAACCTGGTCCGTCAG-1
```

protospacer\_umi\_thresholds.csv

```
Protospacer,UMI threshold
sgRNA_1,6
sgRNA_10,7
sgRNA_100,4
sgRNA_11,10
sgRNA_12,7
sgRNA_13,52
sgRNA_14,2
sgRNA_15,11
sgRNA_16,3
```

matrix.mtx.gz

```
%%MatrixMarket matrix
%metadata_json: {"so
36701 7237 38830775
25 1 3
30 1 16
45 1 3
46 1 3
49 1 2
53 1 1
54 1 2
58 1 1
60 1 8
```

features.tsv.gz

```
ENSG00000278633 AC023491.2 Gene
ENSG00000276017 AC007325.1 Gene
ENSG00000278817 AC007325.4 Gene
ENSG00000277196 AC007325.2 Gene
sgRNA_1 sgRNA_1 CRISPR Guide Capture
sgRNA_2 sgRNA_2 CRISPR Guide Capture
sgRNA_3 sgRNA_3 CRISPR Guide Capture
sgRNA_4 sgRNA_4 CRISPR Guide Capture
sgRNA_5 sgRNA_5 CRISPR Guide Capture
sgRNA_6 sgRNA_6 CRISPR Guide Capture
```



## 3.4.1 数据准备

```
mkdir -p ~/scCRISPR
```

```
cd ~/scCRISPR
```

```
WorkDir=$(pwd)
```

### ### 数据准备

```
mkdir ${WorkDir}/01.CellRanger
```

```
cd ${WorkDir}/01.CellRanger
```

```
ln -sf /CRISPR/scCRISPR/01.CellRanger/barcodes.tsv.gz
```

```
ln -sf /CRISPR/scCRISPR/01.CellRanger/features.tsv.gz
```

```
ln -sf /CRISPR/scCRISPR/01.CellRanger/matrix.mtx.gz
```

```
ln -sf
```

```
/CRISPR/scCRISPR/01.CellRanger/scCRISPR.RawMatrix.txt
```

```
cd ${WorkDir}
```

```
Aug 15 18:37 barcodes.tsv.gz -> /CRISPR/scCRISPR/01.CellRanger/barcodes.tsv.gz
Aug 15 18:37 features.tsv.gz -> /CRISPR/scCRISPR/01.CellRanger/features.tsv.gz
Aug 15 18:37 matrix.mtx.gz -> /CRISPR/scCRISPR/01.CellRanger/matrix.mtx.gz
Aug 15 18:37 scCRISPR.RawMatrix.txt -> /CRISPR/scCRISPR/01.CellRanger/scCRISPR.RawMatrix.txt
```

| Cells              | gRNA | TargetGene                                 | gRNAsInCell | GenesInCell |
|--------------------|------|--------------------------------------------|-------------|-------------|
| AAACCTGAGACTTTCG-1 |      | gRNA48; Nkx3-1; 1                          | 2699        |             |
| AAACCTGAGAGACGAA-1 |      | gRNA73; Agmat; 1                           | 4498        |             |
| AAACCTGAGATATGGT-1 |      | gRNA11;gRNA100; D1Ert622e;NCsgRNA-1; 2     |             |             |
| AAACCTGAGATCGATA-1 |      | gRNA16;gRNA81; Tyro3;Tfap4; 2              | 4903        |             |
| AAACCTGAGATGTAAC-1 |      | gRNA47; Nkx3-1; 1                          | 3651        |             |
| AAACCTGAGCACCGTC-1 |      | - - 0                                      | 2938        |             |
| AAACCTGAGCCGATTT-1 |      | gRNA5;gRNA39;gRNA98; Shf;Tnfrsf1b;Dnajb7;  |             |             |
| AAACCTGAGCGTAATA-1 |      | gRNA5;gRNA62; Shf;Cplx3; 2                 | 5660        |             |
| AAACCTGAGCGTGTCC-1 |      | gRNA37; Tnfrsf1b; 1                        | 2971        |             |
| AAACCTGAGCTCTCGG-1 |      | gRNA98; Dnajb7; 1                          | 3596        |             |
| AAACCTGAGCTGAACG-1 |      | gRNA79; Tfap4; 1                           | 3381        |             |
| AAACCTGAGGCTAGGT-1 |      | gRNA4;gRNA26;gRNA27;gRNA33;gRNA37;gRNA83;  |             |             |
| AAACCTGAGGCTCAGA-1 |      | gRNA2; Adipor2; 1                          | 3830        |             |
| AAACCTGAGGTAGCTG-1 |      | gRNA21; Opnlsw; 1                          | 5004        |             |
| AAACCTGAGTGATCGG-1 |      | gRNA5;gRNA18;gRNA103; Shf;Tyro3;NCsgRNA-4; |             |             |
| AAACCTGAGTTCCACA-1 |      | - - 0                                      | 4097        |             |
| AAACCTGAGTTGTAGA-1 |      | gRNA101; NCsgRNA-2; 1                      | 3911        |             |
| AAACCTGCAAACGCGA-1 |      | gRNA91;gRNA108; 2010107G23Rik;NCsgRNA-9;   |             |             |
| AAACCTGCAAAGCAAT-1 |      | gRNA74; Agmat; 1                           | 3935        |             |
| AAACCTGCAAGAGGCT-1 |      | gRNA89; Tbc1d30; 1                         | 4433        |             |
| AAACCTGCAAGGCTCC-1 |      | gRNA88; Tbc1d30; 1                         | 4502        |             |
| AAACCTGCACATAACC-1 |      | gRNA12; D1Ert622e; 1                       | 1968        |             |
| AAACCTGCACCTATCC-1 |      | gRNA31; Ccdc129; 1                         | 2948        |             |
| AAACCTGCACGGTTTA-1 |      | gRNA84; Elovl1; 1                          | 2979        |             |
| AAACCTGCACTTAAGC-1 |      | gRNA12; D1Ert622e; 1                       | 3323        |             |
| AAACCTGCAGACGCAA-1 |      | gRNA91; 2010107G23Rik; 1                   | 2960        |             |
| AAACCTGCAGAGTGTG-1 |      | gRNA99; Dnajb7; 1                          | 4269        |             |
| AAACCTGCAGCTCGAC-1 |      | gRNA39; Tnfrsf1b; 1                        | 3274        |             |

## 3.4.2 转染效率可视化

### ### 转染效率可视化

```
mkdir -p ${WorkDir}/02.MOIVisualization
cd ${WorkDir}/02.MOIVisualization
awk -F "\t" 'NR>1{if($4==0){print $1 "\tEmpty"}\
else if($4==1){print $1 "\tLMOI"}\
else{print $1 "\tHMOI"}}' \
${WorkDir}/01.CellRanger/scCRISPR.RawMatrix.txt > Visualization.txt
```

| Cells               | gRNA                                                     | TargetGene | gRNAsInCell | GenesInCell |
|---------------------|----------------------------------------------------------|------------|-------------|-------------|
| AAACCTGAGACTTTTCG-1 | gRNA48; NKx3-1; 1                                        |            | 2699        |             |
| AAACCTGAGAGACGAA-1  | gRNA73; Agmat; 1                                         |            | 4498        |             |
| AAACCTGAGATATGGT-1  | gRNA11;gRNA100; D1Ert622e;NCsgRNA-1; 2                   |            | 4730        |             |
| AAACCTGAGATCGATA-1  | gRNA16;gRNA81; Tyro3;Tfap4; 2                            |            | 4903        |             |
| AAACCTGAGATGTAAC-1  | gRNA47; Nkx3-1; 1                                        |            | 3651        |             |
| AAACCTGAGCACCCTC-1  | -                                                        |            | 2938        |             |
| AAACCTGAGCCGATTT-1  | gRNA5;gRNA39;gRNA98; Shf;Tnfrsf1b;Dnajb7; 3              |            | 3801        |             |
| AAACCTGAGCGTAATA-1  | gRNA5;gRNA62; Shf;Cplx3; 2                               |            | 5660        |             |
| AAACCTGAGCGTGTCC-1  | gRNA37; Tnfrsf1b; 1                                      |            | 2971        |             |
| AAACCTGAGCTCTCGG-1  | gRNA98; Dnajb7; 1                                        |            | 3596        |             |
| AAACCTGAGCTGAACG-1  | gRNA79; Tfap4; 1                                         |            | 3381        |             |
| AAACCTGAGGCTAGGT-1  | gRNA4;gRNA26;gRNA27;gRNA33;gRNA37;gRNA83; Shf;Gabrg2;Gal |            |             |             |
| AAACCTGAGGCTCAGA-1  | gRNA2; Adipor2; 1                                        |            | 3830        |             |
| AAACCTGAGGTAGCTG-1  | gRNA21; Opnlsw; 1                                        |            | 5004        |             |
| AAACCTGAGTGATCGG-1  | gRNA5;gRNA18;gRNA103; Shf;Tyro3;NCsgRNA-4; 3             |            | 5044        |             |
| AAACCTGAGTTCCACA-1  | -                                                        |            | 4097        |             |
| AAACCTGAGTTGTAGA-1  | gRNA101; NCsgRNA-2; 1                                    |            | 3911        |             |
| AAACCTGCAACGCGA-1   | gRNA91;gRNA108; 2010107G23Rik;NCsgRNA-9; 2               |            | 4289        |             |
| AAACCTGCAAGCAAT-1   | gRNA74; Agmat; 1                                         |            | 3935        |             |
| AAACCTGCAAGAGGCT-1  | gRNA89; Tbc1d30; 1                                       |            | 4433        |             |
| AAACCTGCAAGGCTCC-1  | gRNA88; Tbc1d30; 1                                       |            | 4502        |             |
| AAACCTGCACATAACC-1  | gRNA12; D1Ert622e; 1                                     |            | 1968        |             |
| AAACCTGCACCTATCC-1  | gRNA31; Ccdc129; 1                                       |            | 2948        |             |
| AAACCTGCACGGTTTA-1  | gRNA84; Elovl1; 1                                        |            | 2979        |             |
| AAACCTGCACCTAACC-1  | gRNA12; D1Ert622e; 1                                     |            | 2222        |             |



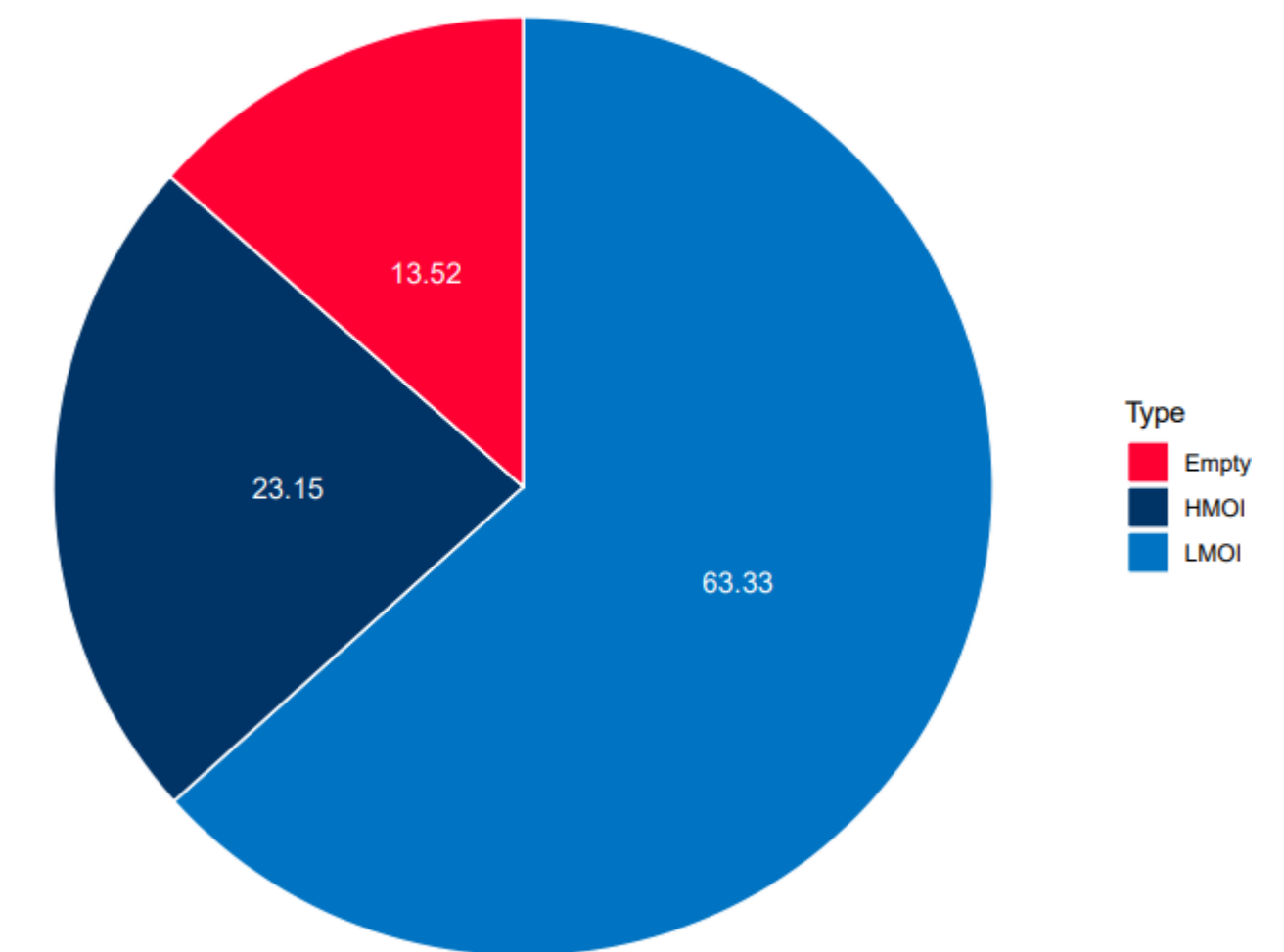
|                     |       |
|---------------------|-------|
| AAACCTGAGACTTTTCG-1 | LMOI  |
| AAACCTGAGAGACGAA-1  | LMOI  |
| AAACCTGAGATATGGT-1  | HMOI  |
| AAACCTGAGATCGATA-1  | HMOI  |
| AAACCTGAGATGTAAC-1  | LMOI  |
| AAACCTGAGCACCCTC-1  | Empty |
| AAACCTGAGCCGATTT-1  | HMOI  |
| AAACCTGAGCGTAATA-1  | HMOI  |
| AAACCTGAGCGTGTCC-1  | LMOI  |
| AAACCTGAGCTCTCGG-1  | LMOI  |
| AAACCTGAGCTGAACG-1  | LMOI  |
| AAACCTGAGGCTAGGT-1  | HMOI  |
| AAACCTGAGGCTCAGA-1  | LMOI  |
| AAACCTGAGGTAGCTG-1  | LMOI  |
| AAACCTGAGTGATCGG-1  | HMOI  |
| AAACCTGAGTTCCACA-1  | Empty |
| AAACCTGAGTTGTAGA-1  | LMOI  |
| AAACCTGCAACGCGA-1   | HMOI  |
| AAACCTGCAAGCAAT-1   | LMOI  |
| AAACCTGCAAGAGGCT-1  | LMOI  |
| AAACCTGCAAGGCTCC-1  | LMOI  |
| AAACCTGCACATAACC-1  | LMOI  |
| AAACCTGCACCTATCC-1  | LMOI  |

## 3.4.2 转染效率可视化

```
### 这一段为 R 代码需要先在命令行输入 R + 回车 进入R界面
library("ggplot2")
library("dplyr")
Input <- "Visualization.txt"
Output <- "MOI.pdf"
mydata <- read.table(Input,sep="\t",row.names=1)
mydata <- as.data.frame(table(mydata))
mydata_sum <- sum(mydata[,2])
mydata[,2] <- round(100*mydata[,2]/mydata_sum,2)
colnames(mydata)<-c("Type","Freq")
mydata <- mydata %>% arrange(desc(Type)) %>% mutate(pos = cumsum(Freq) - 0.5*Freq)
mycols <- c("#FF0033FF","#003366FF","#0073C2FF", "#EFC000FF", "#868686FF", "#CD534CFF")
mycols <- mycols[1:dim(mydata)[1]]

pdf(Output)
ggplot(mydata, aes(x = "", y = Freq, fill = Type))+
  geom_bar(width = 1, stat = "identity", color = "white")+
  coord_polar("y", start = 0)+
  geom_text(aes(y = pos, label = Freq), color = "white")+
  scale_fill_manual(values = mycols)+theme(legend.title=element_blank())+
  theme_void()
dev.off()
q(save="no")
### R 代码结束

cd ${WorkDir}
```





## 3.4.3 过滤高MOI细胞

```
%%MatrixMarket matrix coordinate integer general
%metadata_json: {"software_version": "cellranger-6.0.1", "format_version": 2}
32394 25764 93264309
9 1 3
11 1 1
35 1 1
54 1 1
64 1 14
70 1 2
97 1 1
132 1 1
145 1 1
146 1 1
150 1 1
```



```
%%MatrixMarket matrix coordinate integer general
%metadata_json: {"software_version": "cellranger-6.0.1", "format_version": 2}
32394 19800 71584647
9 1 1
11 1 1
14 1 1
17 1 2
```

```
### 过滤高 MOI 细胞
mkdir -p ${WorkDir}/03.Filter
cd ${WorkDir}/03.Filter
inputdir=${WorkDir}/01.CellRanger
outputdir=${WorkDir}/03.Filter
RawMatrix=${WorkDir}/01.CellRanger/scCRISPR.RawMatrix.txt
cd ${outputdir}
cp ${inputdir}/barcodes.tsv.gz ${outputdir}
cp ${inputdir}/features.tsv.gz ${outputdir}
cp ${inputdir}/matrix.mtx.gz ${outputdir}
### 筛出低MOI细胞的顺序号
awk -F "\t" '$4<=1{print NR}' ${RawMatrix} > LMOI_Cells_ID.txt
### 重构 matrix.mtx.gz
gunzip matrix.mtx.gz
head -n 2 matrix.mtx > matrix.filtered.mtx.head
awk 'ARGIND==1{a[$1]=FNR} ARGIND==2{if((FNR>3)&&($2 in a)){print $1 " " a[$2] " "
$3}}' LMOI_Cells_ID.txt matrix.mtx > matrix.filtered.mtx
gene_count=`zcat features.tsv.gz |wc -l`
cell_count=`awk '{print $2}' matrix.filtered.mtx |uniq|wc -l`
item_count=`cat matrix.filtered.mtx|wc -l`
echo "${gene_count} ${cell_count} ${item_count}" >> matrix.filtered.mtx.head
cat matrix.filtered.mtx.head matrix.filtered.mtx > matrix.mtx
gzip matrix.mtx
### 重构 barcodes.tsv.gz
gunzip barcodes.tsv.gz
awk 'ARGIND==1{a[$1]} ARGIND==2&&(FNR in a){print}' LMOI_Cells_ID.txt barcodes.tsv
> barcodes.filtered.tsv
mv barcodes.filtered.tsv barcodes.tsv
gzip barcodes.tsv
rm -f LMOI_Cells_ID.txt matrix.filtered.mtx matrix.filtered.mtx.head

cd ${WorkDir}
```



## 3.5.1 使用Seurat 读取数据

### ### Seurat 读取数据

```
mkdir -p ${WorkDir}/04.Seurat
cd ${WorkDir}/04.Seurat
```

```
> CRISPR.data <- Read10X(data.dir = "~/scCRISPR/03.Filter/")
10X data contains more than one type and is being returned as a list containing matrices of each type.
```

### 这一段为 R 代码需要先在命令行输入 R + 回车 进入R界面

```
library(Seurat)
library(ggplot2)
library(dplyr)
library(patchwork)
```

### # 导入数据

```
CRISPR.data <- Read10X(data.dir = "~/scCRISPR/03.Filter/")
```

### ### 查看第一个list

```
head(CRISPR.data[[1]],n=5)[,1:5]
```

### ### 查看第二个list

```
head(CRISPR.data[[2]],n=5)[,1:5]
```

```
CRISPR.obj <- CreateSeuratObject(counts = CRISPR.data[[1]],
project = "scCRISPR", min.cells = 0, min.features = 200)
```

```
> head(CRISPR.data[[1]],n=5)[,1:5]
head(CRISPR.data[[2]],n=5)[,1:5]
5 x 5 sparse Matrix of class "dgCMatix"
      AACCTGAGAGACGAA-1 AACCTGAGATATGGT-1 AACCTGAGCACCCTC-1
Xkr4      .              .              .              .
Gm1992      .              .              .              .
Gm19938      .              .              .              .
Gm37381      .              .              .              .
Rp1         .              .              .              .
      AACCTGAGCCGATTT-1 AACCTGAGCTCTCGG-1
Xkr4      .              .              .              .
Gm1992      .              .              .              .
Gm19938      .              .              .              .
Gm37381      .              .              .              .
Rp1         .              .              .              .
5 x 5 sparse Matrix of class "dgCMatix"
      AACCTGAGAGACGAA-1 AACCTGAGATATGGT-1 AACCTGAGCACCCTC-1
gRNA1      .              .              .              .
gRNA2      1              .              .              1
gRNA3      .              .              .              .
gRNA4      .              1              .              2
gRNA5      .              .              .              .
      AACCTGAGCCGATTT-1 AACCTGAGCTCTCGG-1
gRNA1      .              1              .              .
gRNA2      .              .              .              .
gRNA3      .              2              .              .
gRNA4      .              .              .              .
gRNA5      12             .              .              .
```



## 3.5.2 Seurat 基本质控

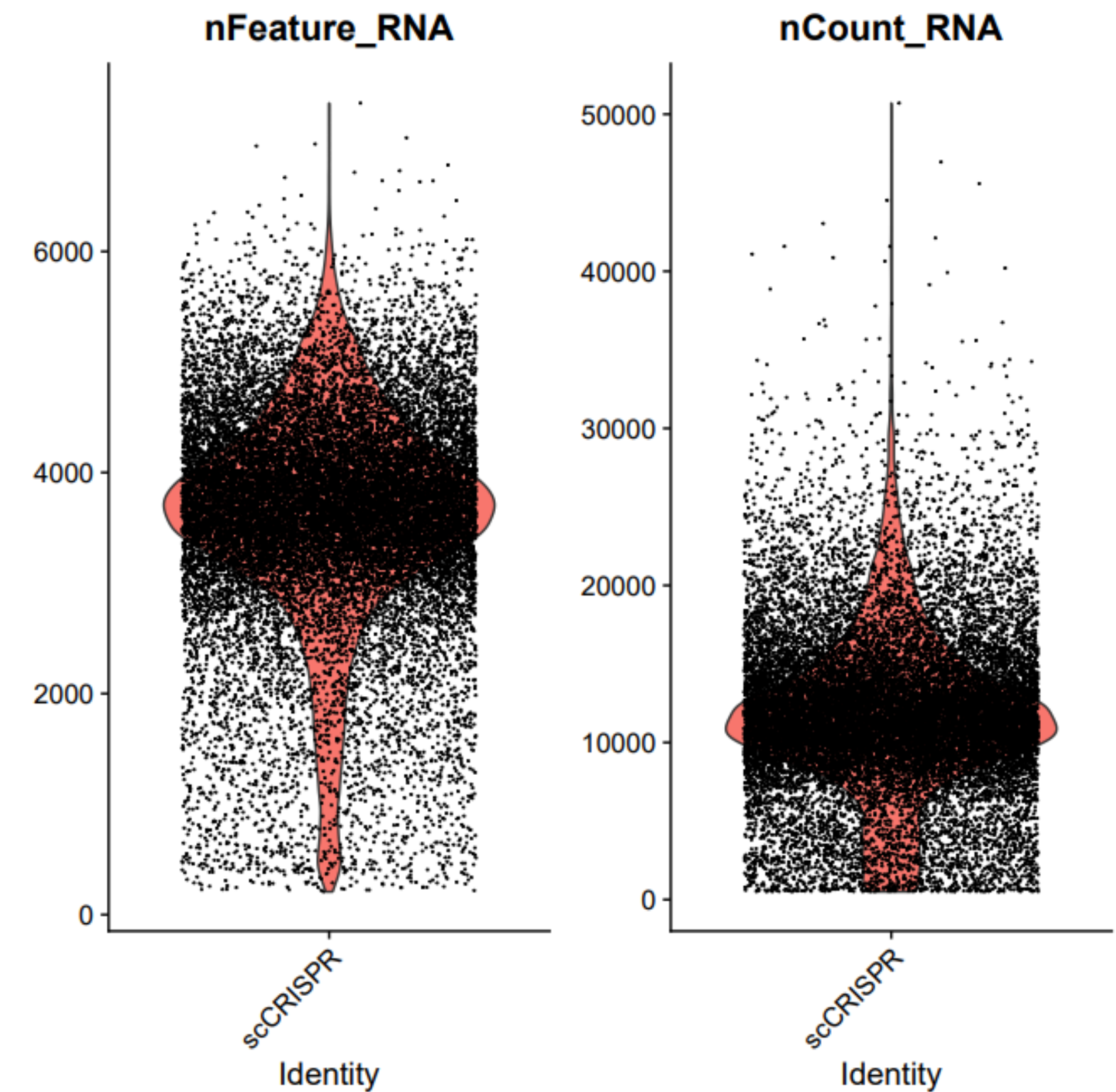
### # Seurat 质控

### 先作图，根据图定阈值

```
pdf("QC.pdf")  
# CRISPR.obj[["percent.mt"]] <-  
# PercentageFeatureSet(CRISPR.obj, pattern = "^MT-")  
VlnPlot(CRISPR.obj, features = c("nFeature_RNA", "nCount_RNA"), ncol = 2)  
dev.off()
```

### 根据右图手动定过滤标准

```
CRISPR.obj <- subset(CRISPR.obj,  
subset = nFeature_RNA > 200 & nFeature_RNA < 7000 & nCount_RNA  
< 35000)
```



# 3.5.3 数据标准化与高变基因的寻找

## 标准化 (Normalization)

目的：消除测序深度差异带来的影响

方法：除以每个细胞总读数并乘以一个标准化因子

## # 数据标准化

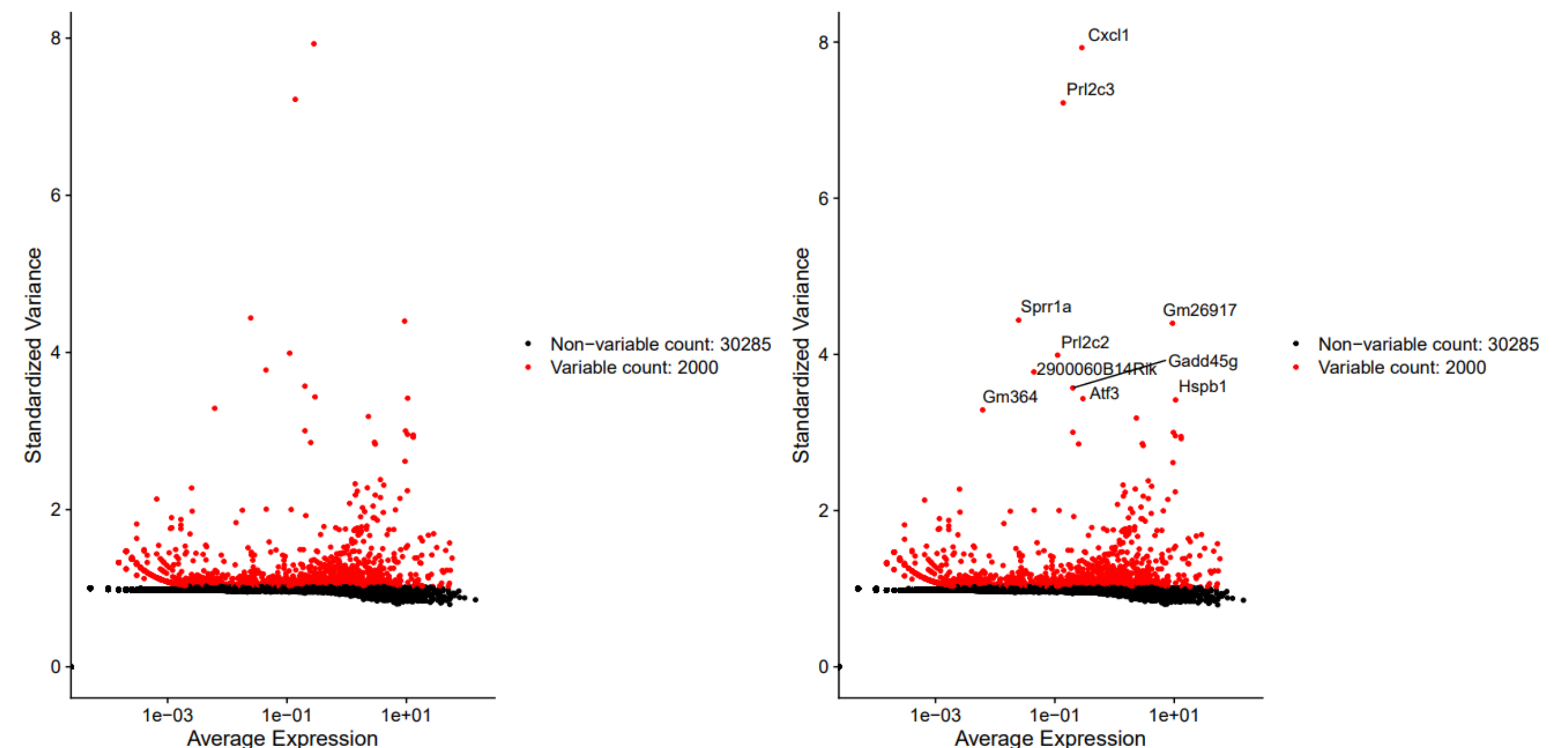
```
CRISPR.obj <- NormalizeData(object = CRISPR.obj,  
  normalization.method = "LogNormalize", scale.factor =  
  10000)
```

## # 寻找高变基因

```
CRISPR.obj <- FindVariableFeatures(CRISPR.obj,  
  selection.method = "vst", nfeatures = 2000)
```

## # 可视化高变基因

```
top10 <- head(VariableFeatures(CRISPR.obj), 10)  
plot1 <- VariableFeaturePlot(CRISPR.obj)  
plot2 <- LabelPoints(plot = plot1, points = top10, repel =  
  TRUE)  
pdf("VariableFeatures.pdf",width=14)  
plot1 + plot2  
dev.off()
```



## 3.5.4 数据归一化

### 归一化 (Scaling)

目的：使数据适合于下游的机器学习和统计分析（使表达量分布符合统计学分布）

方法：减去平均值并除以标准差使矩阵每个基因表达均值为0，所有基因的表达分布基本一致（正态分布）

# 数据归一化

```
CRISPR.genes <- rownames(CRISPR.obj)
```

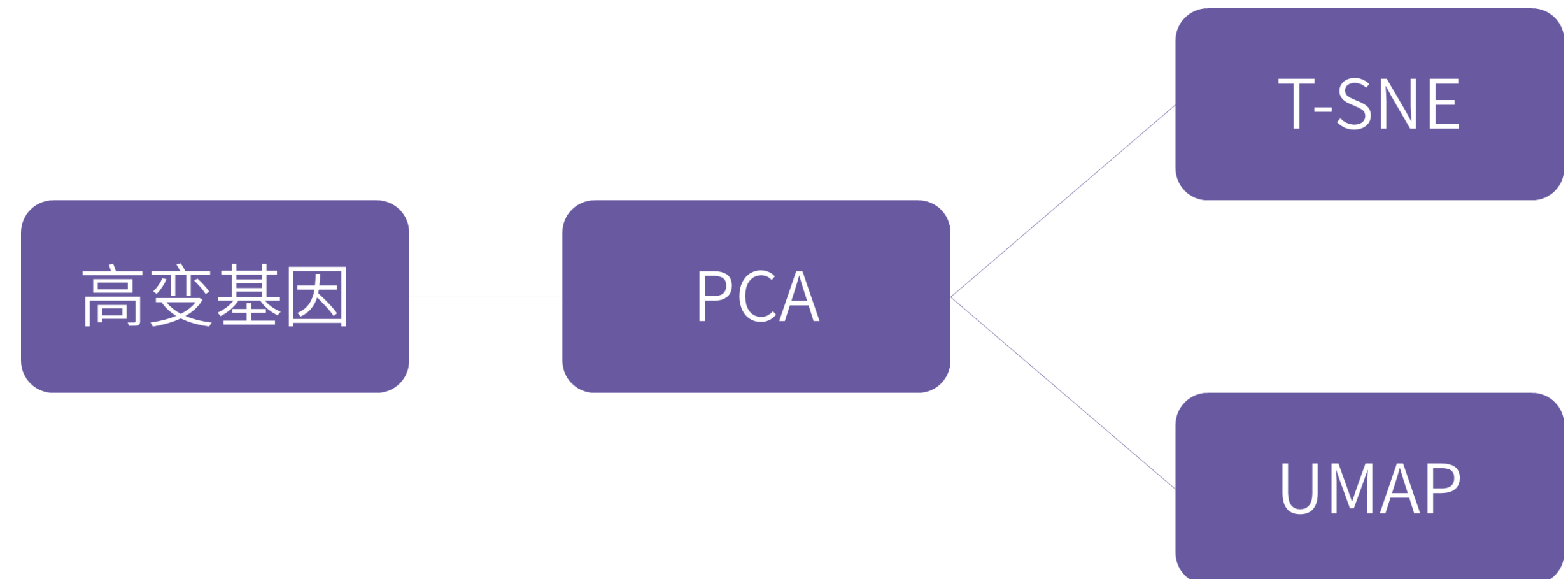
```
CRISPR.obj <- ScaleData(CRISPR.obj, features = CRISPR.genes)
```



## 3.5.5.1 数据降维与聚类

1. 维度灾难
2. 过拟合问题
3. 计算复杂性高
4. 难以可视化

解决方案：在保留信息的同时将维度数量降下来。



聚类分析是量化计算特征相似度,将功能相同或密切联系的细胞聚集成类。



## 3.5.5.2 主成分分析 (PCA)

### # 运行PCA

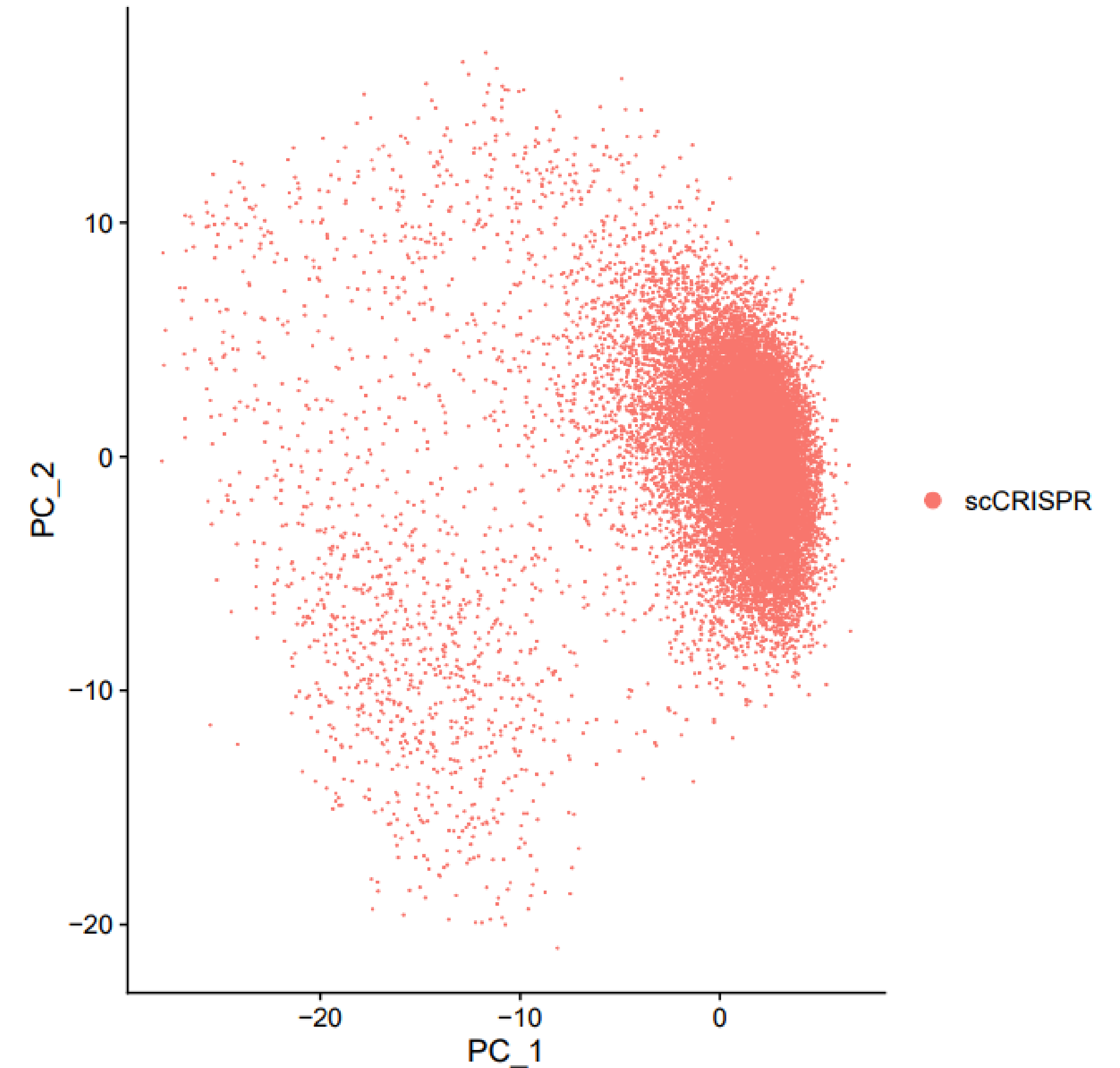
```
CRISPR.obj <- RunPCA(CRISPR.obj, features = VariableFeatures(object =  
CRISPR.obj))
```

### ### 查看pca结果

```
pdf("PCA.pdf")  
DimPlot(CRISPR.obj, reduction = "pca")  
dev.off()
```

主成分分析 (PCA, Principal Component Analysis) 是一种常用的数据降维方法，

属于线性降维。它通过线性变换将原始数据变换到一个新的坐标系统中，使得任何数据投影的第一大方差来自坐标系的第一个坐标（称为第一主成分），第二大方差来自第二个坐标（第二主成分），以此类推。





# 3.5.5.3 聚类

## # 获取维数

```
if(dim(CRISPR.obj@reductions$pca[[1,]])[[2]]<100){PCADIMS <- dim(CRISPR.obj@reductions$pca[[1,]])[[2]]}else{PCADIMS <- 100}
```

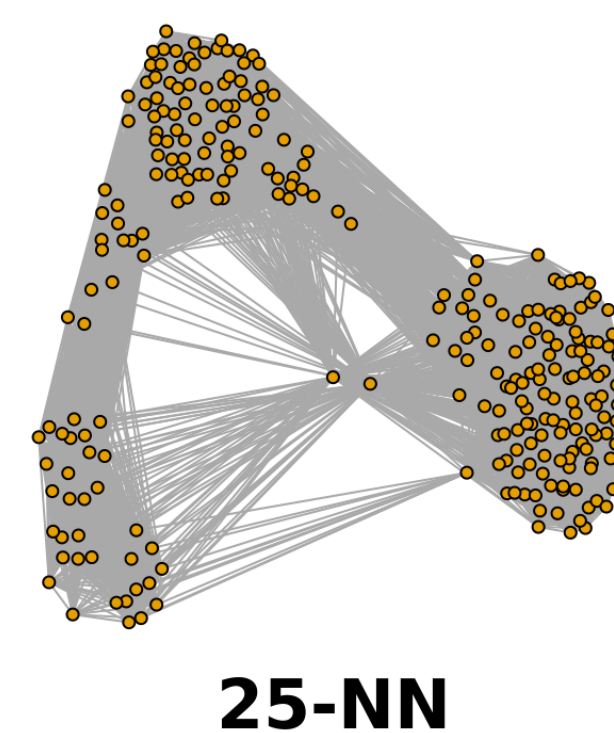
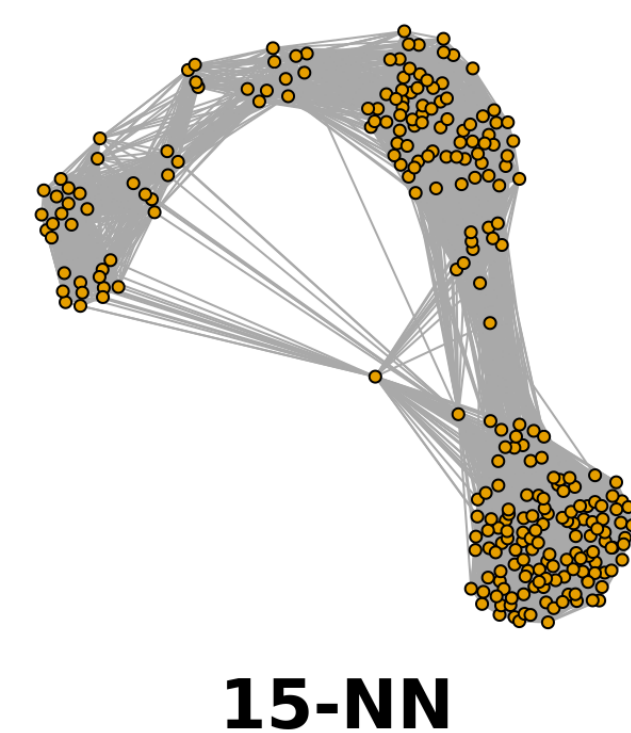
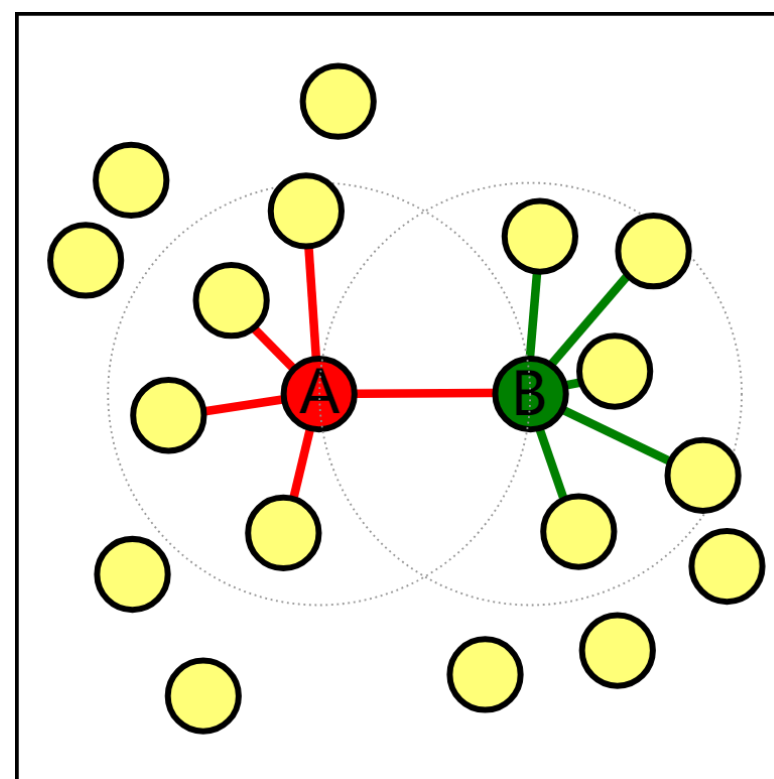
# dims = 1:10和resolution = 0.5可以调整，值越大，聚类越多，根据实际需要不断调整参数，即可得到自己想要的细胞聚类数。

```
CRISPR.obj <- FindNeighbors(CRISPR.obj, dims = 1:PCADIMS)
```

```
CRISPR.obj <- FindClusters(CRISPR.obj, resolution = 0.5)
```

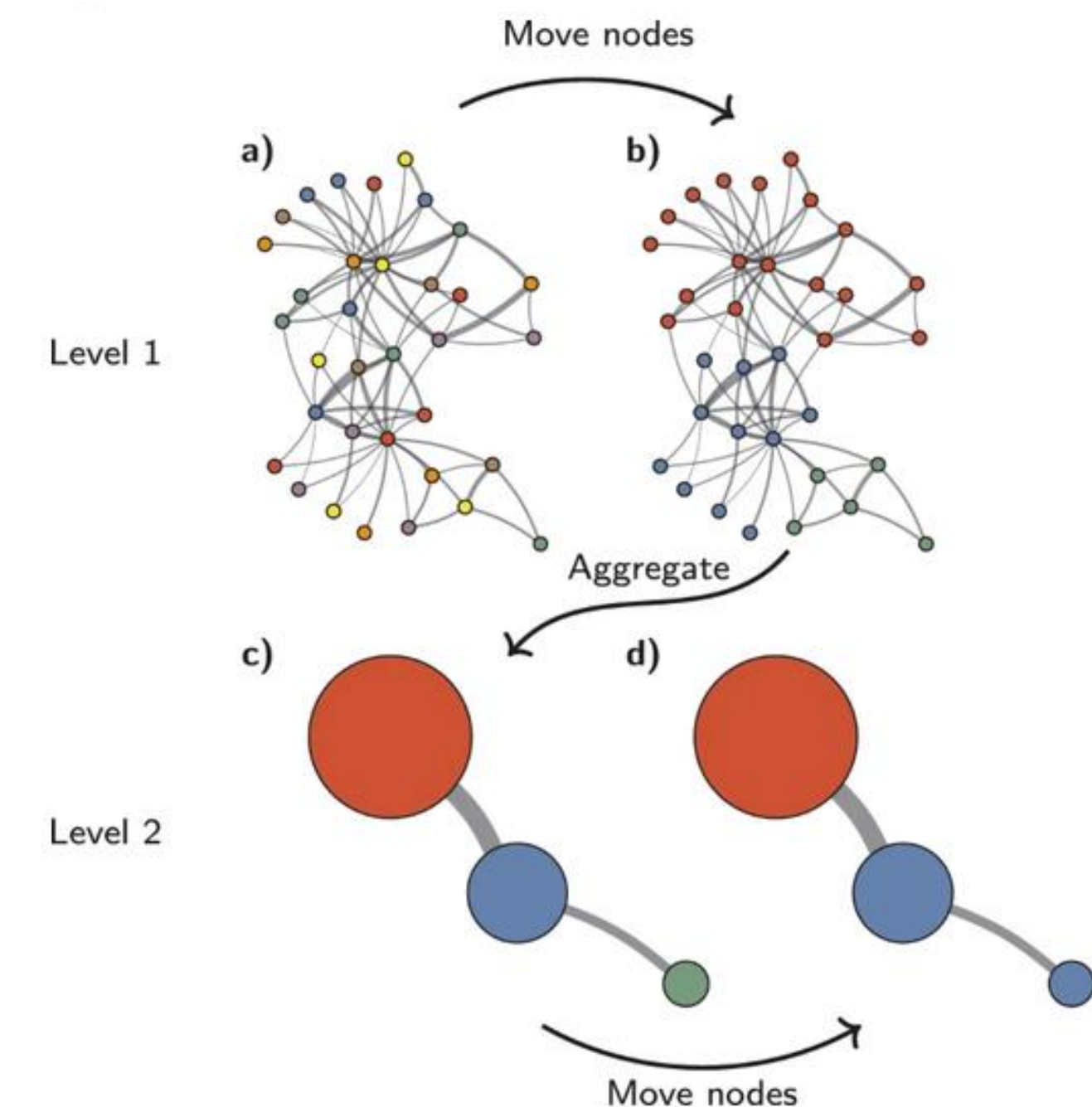
## # # 保存分组信息

```
write.table(as.data.frame(CRISPR.obj$seurat_clusters),"Cluster_Group.txt",quote=F,col.names=F,sep="\t")
```



找到每个细胞其k个最近邻的细胞，然后构建一个k近邻图。在这个图中，每个细胞是一个节点，如果两个细胞是彼此的近邻，那么它们之间就有一条边。

然后将节点划分为不同的社区，并计算新的模块度得分，然后不断迭代直到模块度不再增加为止。

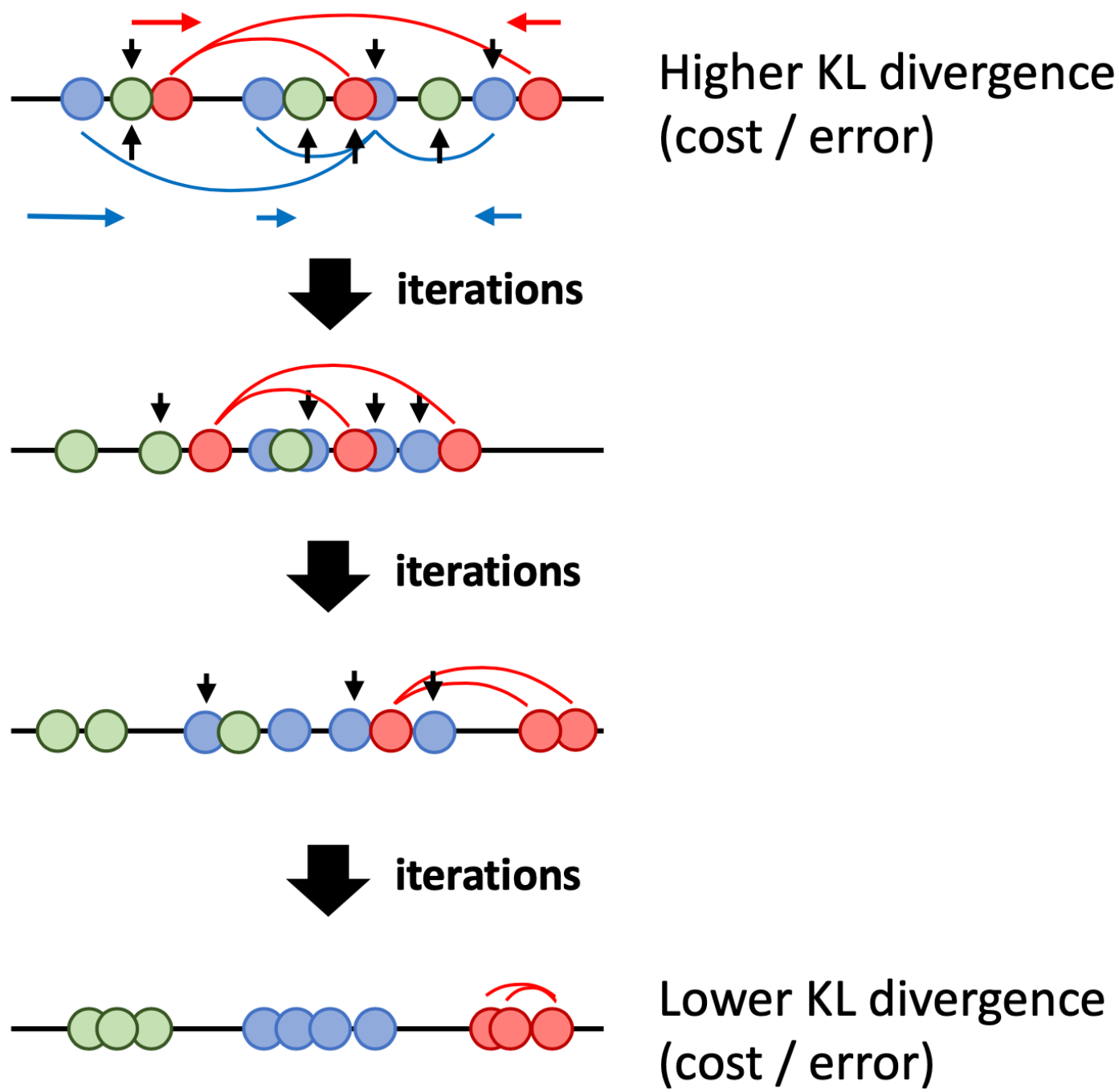
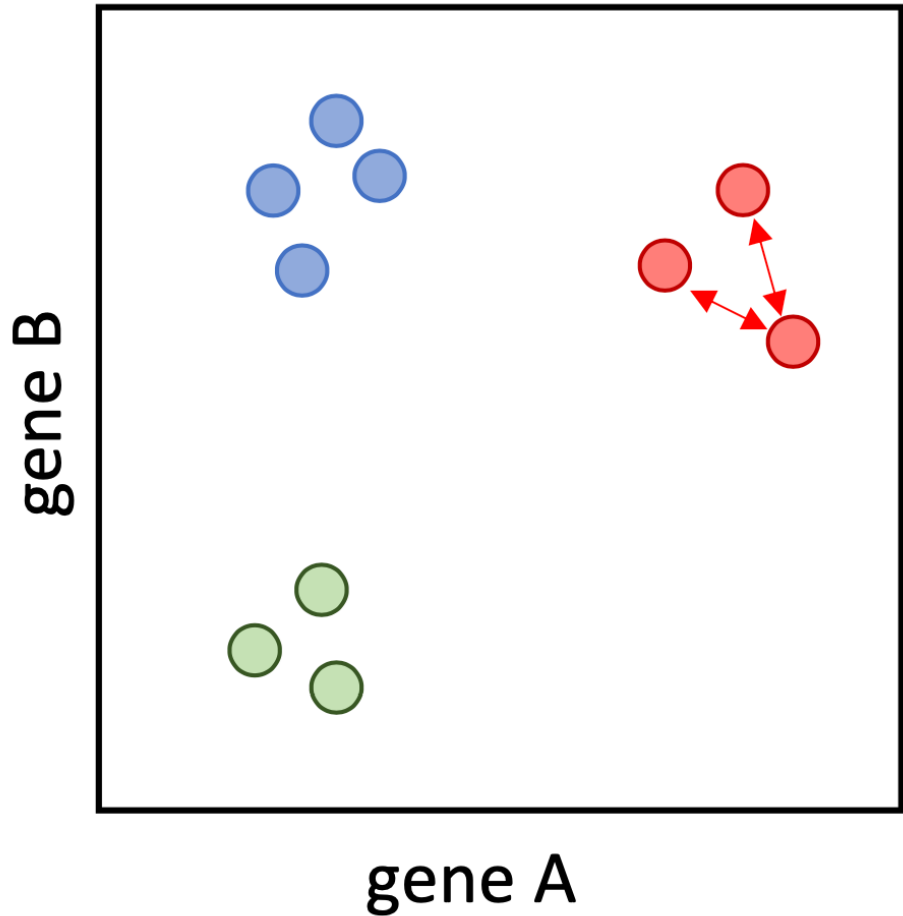


Louvain algorithm. The Louvain algorithm starts from a singleton partition in which each node is in its own community (a). The algorithm moves individual nodes from one community to another to find a partition (b). Based on this partition, an aggregate network is created (c). The algorithm then moves individual nodes in the aggregate network (d). These steps are repeated until the quality cannot be increased further.

# 3.5.6.1 非线性降维之 t-SNE

## 工作原理：

- 1. 计算高维空间中的相似度
- 2. 计算低维空间中的相似度
- 3. 最小化高维空间和低维空间的相似度之间的差异



## 3.5.6.1 非线性降维之 t-SNE

# 若要使用t-SNE聚类

```
CRISPR.obj <- RunTSNE(CRISPR.obj, dims = 1:PCADIMS, check_duplicates=F)
```

```
# DimPlot(CRISPR.obj, reduction = "tsne")
```

# 提取tsne坐标

```
tsne_coordinates <- CRISPR.obj@reductions$tsne[, 1:2]
```

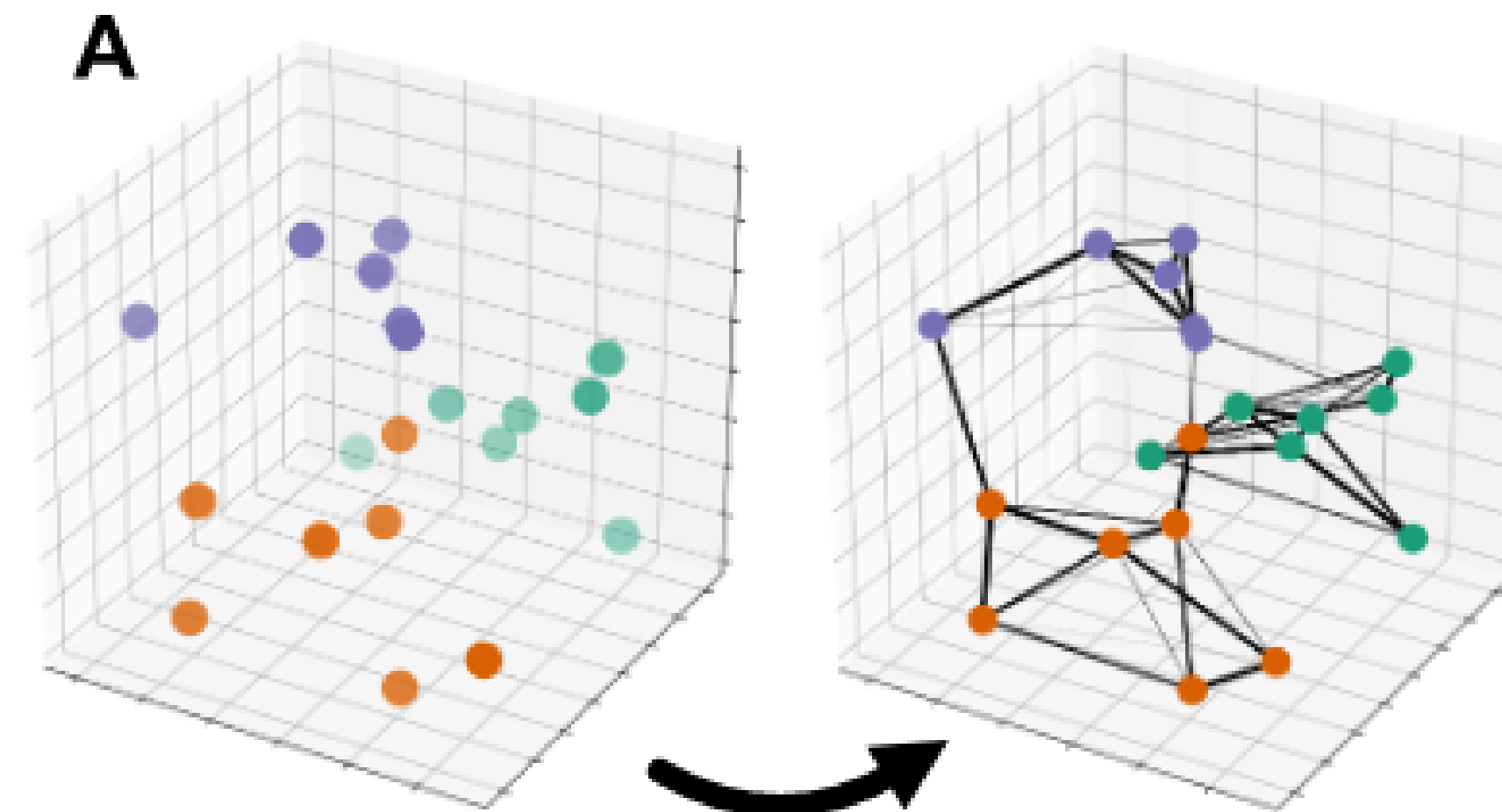
```
write.table(tsne_coordinates, sep = "\t", "tSNE_coord.txt", quote = F, col.names = F)
```



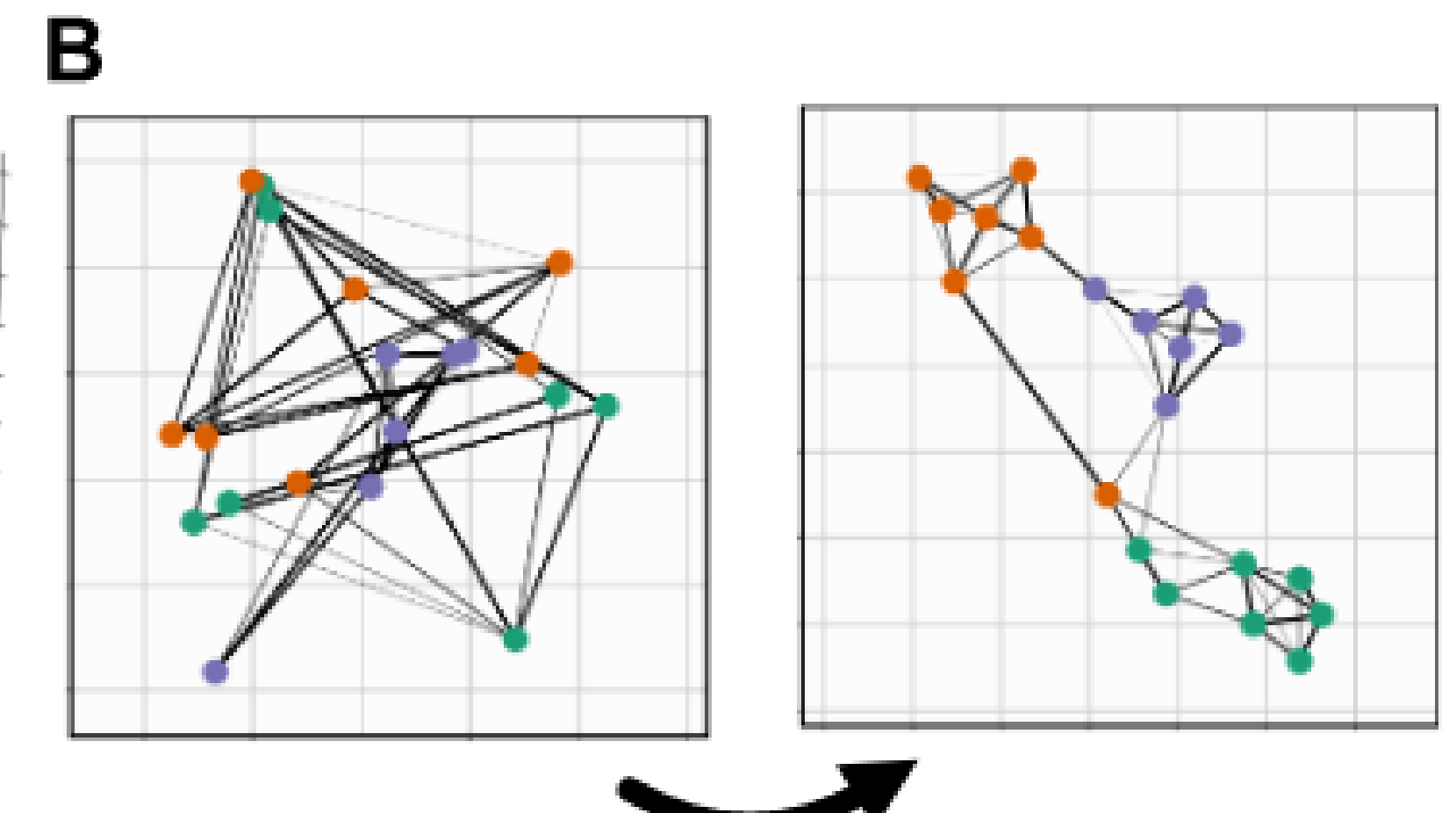
## 3.5.6.2 非线性降维之 UMAP

### 工作原理:

1. 构建高维拓扑
2. 构建低维拓扑
3. 高维拓扑向低维拓扑优化映射



Step 1: Compute a graphical representation of the dataset



Step 2 (non-parametric): Learn an embedding that preserves the structure of the graph



## 3.5.6.2 非线性降维之 UMAP

# 如要使用 UMAP 降维

```
CRISPR.obj <- RunUMAP(CRISPR.obj, dims = 1:PCADIMS)
```

# 提取UMAP坐标

```
umap_coordinates<-CRISPR.obj@reductions$umap[[,1:2]]
```

```
write.table(umap_coordinates,sep="\t","UMAP_coord.txt",quote=F,col.names=F)
```



## 3.5.6.3 数据矩阵的提取与保存



### 保存细胞表达量绝对定量矩阵

```
write.table(as.matrix(CRISPR.obj[["RNA"]]  
@counts),"mRNA_ReadCount.txt",row.names=T,col.names=T,quote=F)
```

### 保存细胞表达量均一化数据矩阵

```
write.table(as.matrix(CRISPR.obj[["RNA"]]  
@data),"mRNA_ScaleCount.txt",row.names=T,col.names=T,quote=F)
```

### 保存细胞表达量标准化数据矩阵

```
write.table(as.matrix(CRISPR.obj[["RNA"]]  
@scale.data),"mRNA_ScaleCount.txt",row.names=T,col.names=T,quote=F)
```

```
q(save="no")
```

## 3.5.6.4 t-SNE 与 UMAP 数据绘图

### # 整理 t-SNE 绘图数据

```
awk -F "\t" 'ARGIND==1{a[$1]=$2} ARGIND==2{b[$1]=$2} \
ARGIND==3{print $0 "\t" a[$1] "\t" b[$1]}' Cluster_Group.txt \
${WorkDir}/02.MOIVisualization/Visualization.txt \
tSNE_coord.txt > plot_tSNE.txt
```

### # 整理 UMAP 绘图数据

```
awk -F "\t" 'ARGIND==1{a[$1]=$2} ARGIND==2{b[$1]=$2} \
ARGIND==3{print $0 "\t" a[$1] "\t" b[$1]}' Cluster_Group.txt \
${WorkDir}/02.MOIVisualization/Visualization.txt \
UMAP_coord.txt > plot_UMAP.txt
```

## 3.5.6.4 t-SNE 与 UMAP 数据绘图

### 这一段为 R 代码需要先在命令行输入 R + 回车 进入R界面

```
library(ggplot2)
mytheme <- theme(plot.title=element_text(face="bold.italic",
size="20", color="black",hjust=0.5,vjust=0.5),
panel.background=element_rect(fill="white",color="black"),
panel.grid.minor.x=element_blank(),legend.position="top",
axis.text = element_blank(),axis.title = element_blank(),
axis.ticks = element_blank())
```

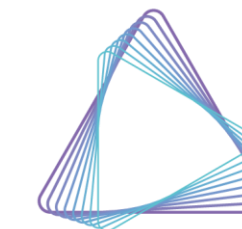
```
mydata<-read.table("plot_tSNE.txt",header=F,sep="\t")
mydata[,4] <- as.character(mydata[,4])
```

### t-SNE 图

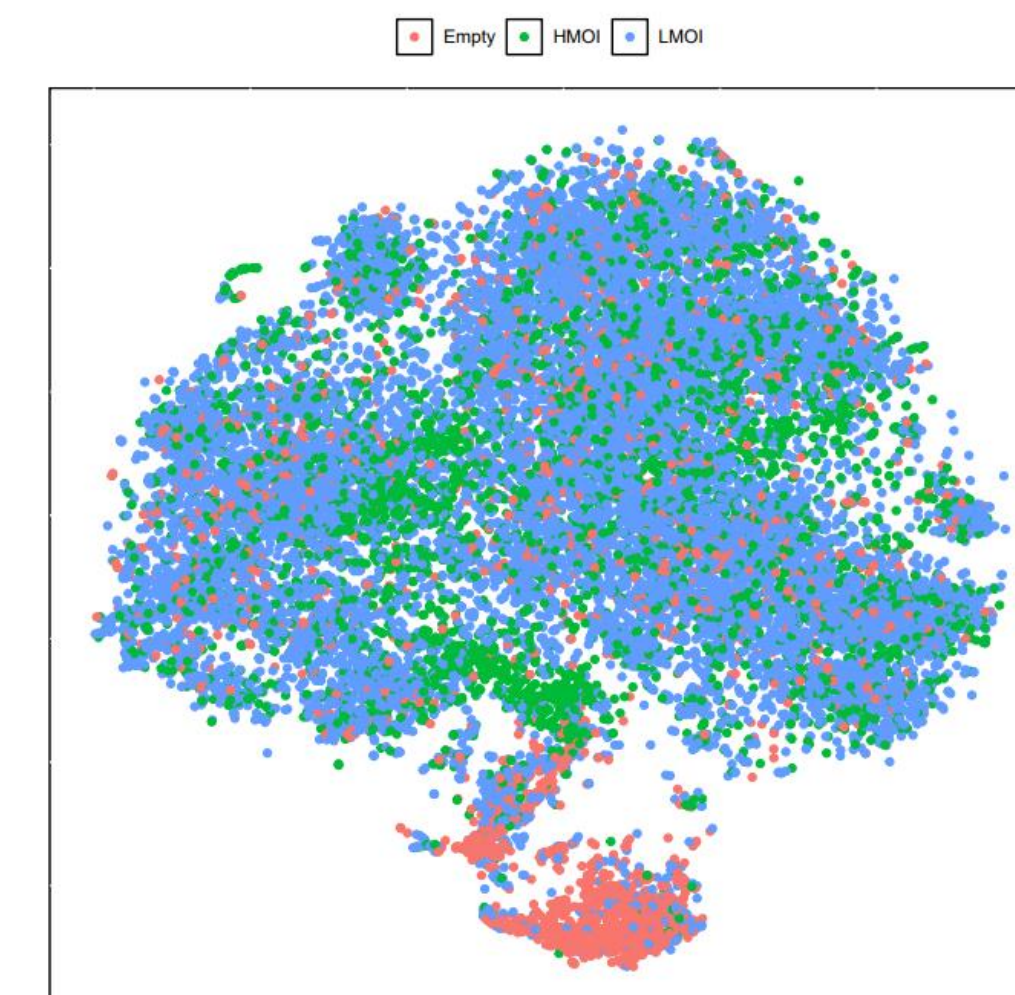
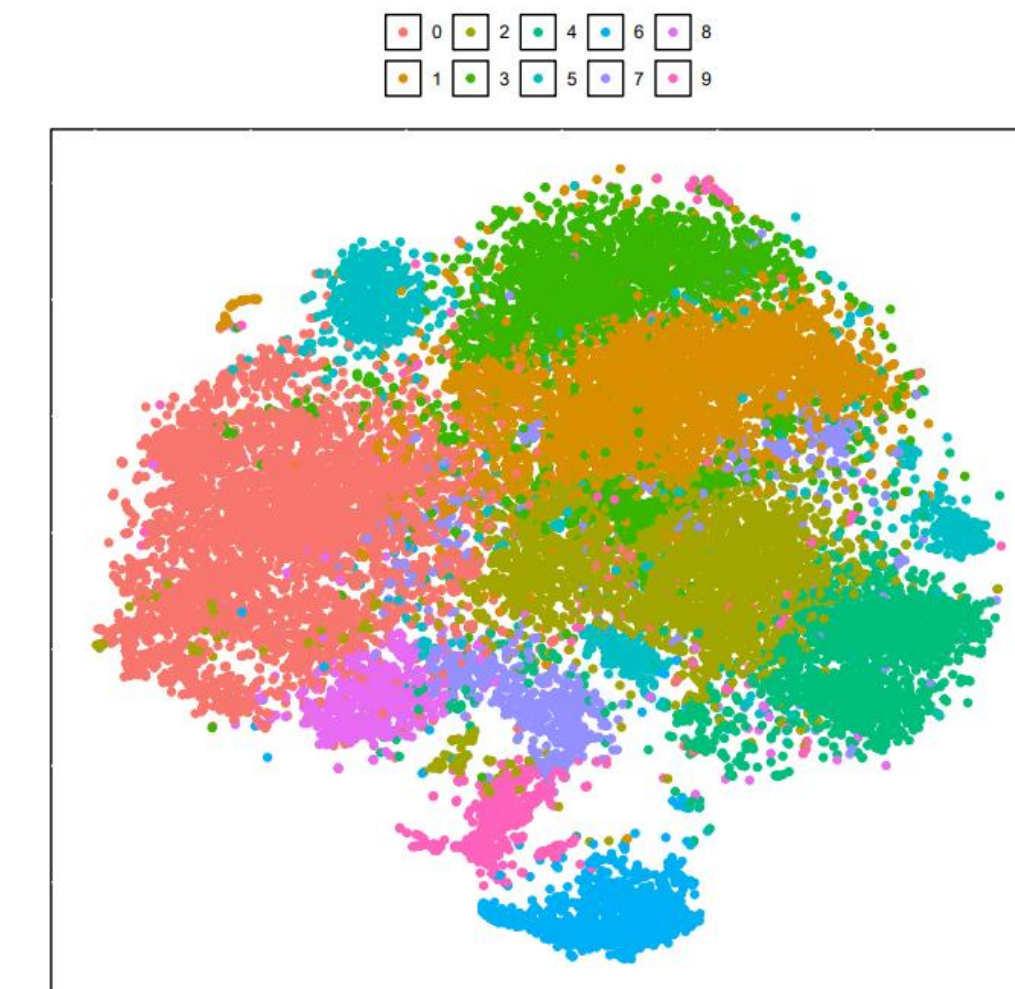
```
pdf("tSNE.pdf",width=7)
ggplot(mydata,aes(x=mydata[,2],y=mydata[,3],color=mydata[,4]))+
geom_point()+
mytheme+xlabs("")+ylabs("")+theme(legend.title=element_blank())
dev.off()
```

### gRNA Guided t-SNE图

```
pdf("tSNE_gRNAGuided.pdf",width=7)
ggplot(mydata,aes(x=mydata[,2],y=mydata[,3],color=mydata[,5]))+
geom_point()+
mytheme+xlabs("")+ylabs("")+theme(legend.title=element_blank())
dev.off()
```



**AZENTA** 安升达  
LIFE SCIENCES





## 3.5.6.4 t-SNE 与 UMAP 数据绘图

### # 读取数据

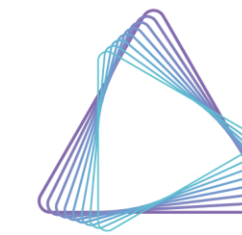
```
mydata<-read.table("plot_UMAP.txt",header=F,sep="\t")
mydata[,4] <- as.character(mydata[,4])
```

### ### t-SNE 图

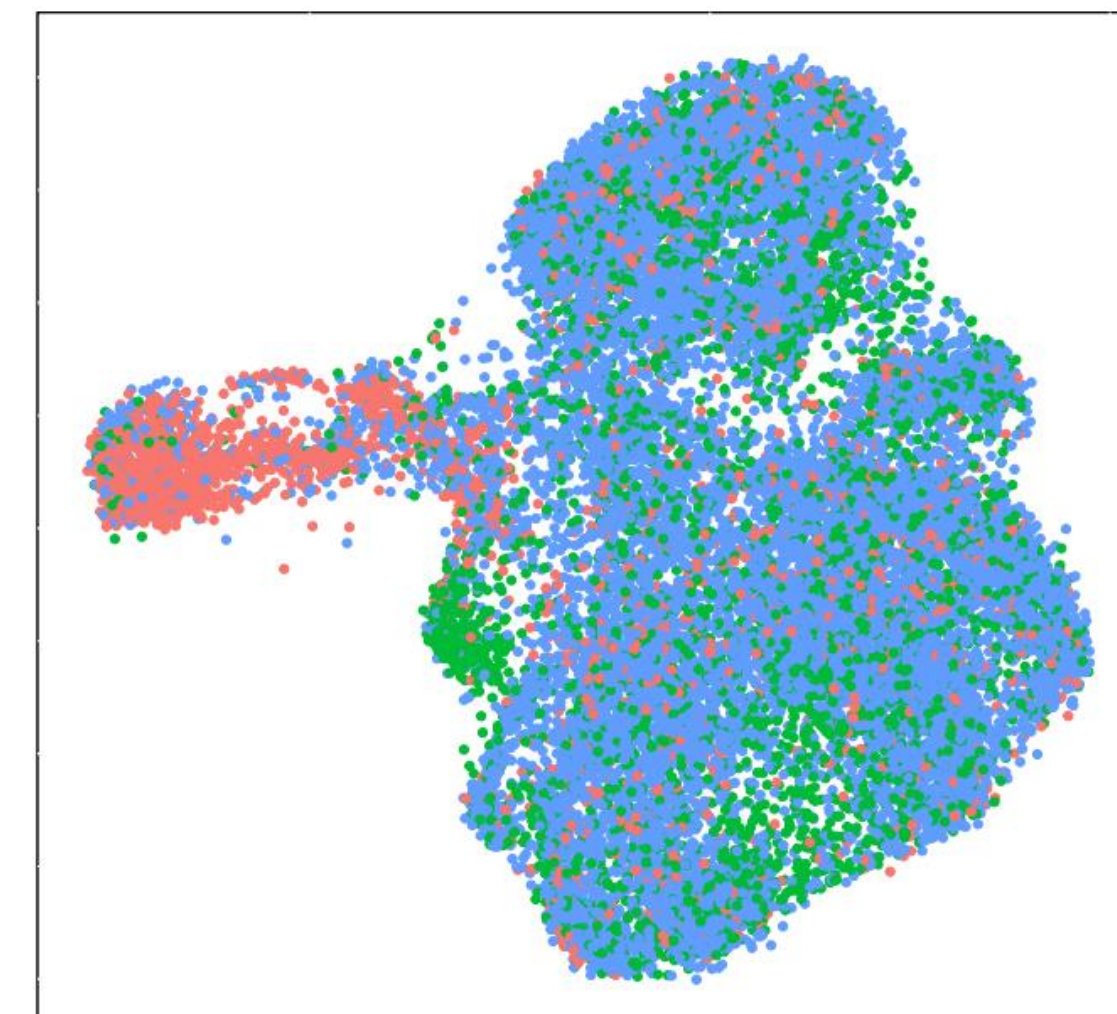
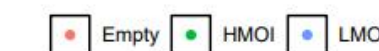
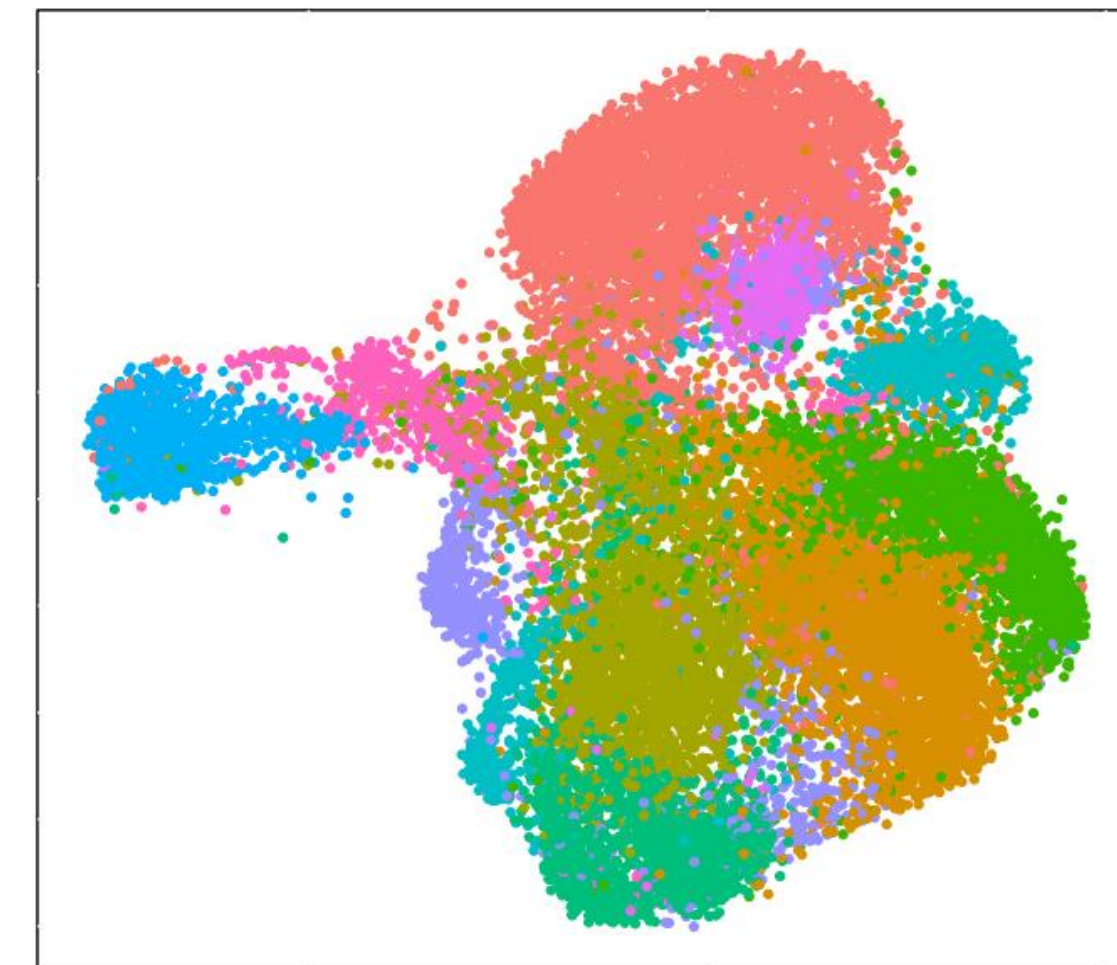
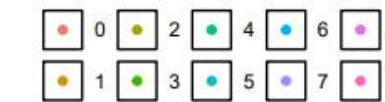
```
pdf("UMAP.pdf",width=7)
ggplot(mydata,aes(x=mydata[,2],y=mydata[,3],color=mydata[,4]))+
geom_point()+
mytheme+xlabs("")+ylabs("")+theme(legend.title=element_blank())
dev.off()
```

### ### gRNA Guided t-SNE图

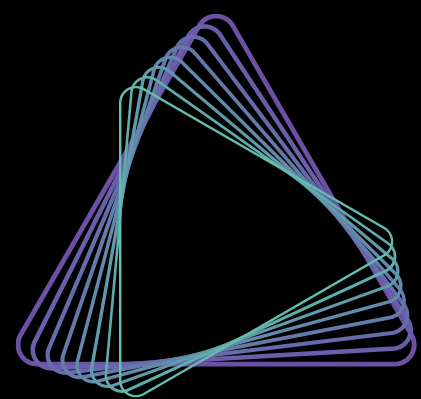
```
pdf("UMAP_gRNAGuided.pdf",width=7)
ggplot(mydata,aes(x=mydata[,2],y=mydata[,3],color=mydata[,5]))+
geom_point()+
mytheme+xlabs("")+ylabs("")+theme(legend.title=element_blank())
dev.off()
```



**AZENTA** 安升达  
LIFE SCIENCES



# Thank you!



**AZENTA 安升达**  
LIFE SCIENCES